Special Relativity: Energy, Momentum, and the Electromagnetic Field

Based on lectures by Leonard Susskind

Lecture notes organized by LazyingArt LLC with Video2Book

Original lecture by Leonard Susskind. Notes reconstructed from the lecture and organized by LazyingArt LLC with Video2Book.

Chapter 1

Energy, Momentum, and the Electromagnetic Field

The final lecture begins with a repair and ends with the object that will stand on the right-hand side of Einstein's equation. The route is not a detour. A sign convention for the electric field must first be made explicit; three distinct meanings of momentum must then be separated. Only after those preliminaries can field theory be read as ordinary mechanics with one degree of freedom at every point in space. The Hamiltonian, local energy density, Noether momentum, electromagnetic energy, and Poynting vector then follow in one continuous argument.

1.1 A Convention That Must Be Carried Consistently

The preliminary notes were to be posted for the class, but the issue is important enough to settle aloud. Byron had noticed an inconsistency not within the course notation, but between that notation and the notation used almost everywhere else. This is an occupational hazard of deriving equations afresh: an internally consistent private convention can drift away from the public one.

Write $\mathbf{E}_M$ for the standard Maxwell–Franklin convention and $\mathbf{E}_L$ for the convention used in the preceding lectures. With the course's covariant component A_0, they are

$$\mathbf{E}_M = -\dot{\mathbf{A}} + \boldsymbol{\nabla} A_0, \qquad \mathbf{E}_L = +\dot{\mathbf{A}} - \boldsymbol{\nabla} A_0 = -\mathbf{E}_M. \tag{1.1}$$

Many books instead introduce the scalar potential $\Phi = -A_0$; then the first formula reads $\mathbf{E}_M = -\dot{\mathbf{A}} - \boldsymbol{\nabla}\Phi$. The magnetic field does not change:

$$\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}. \tag{1.2}$$

In rationalized units with $c = 1$, the standard Maxwell–Franklin equations are

$$\begin{aligned} \boldsymbol{\nabla} \times \mathbf{E}_M &= -\dot{\mathbf{B}}, & \boldsymbol{\nabla} \cdot \mathbf{B} &= 0, \\ \boldsymbol{\nabla} \times \mathbf{B} &= \dot{\mathbf{E}}_M + \mathbf{J}_F, & \boldsymbol{\nabla} \cdot \mathbf{E}_M &= \rho_F. \end{aligned} \tag{1.3}$$

Benjamin Franklin's historical choice fixes the sign of charge: the electron is negative in this convention. The course convention also reverses the source labels,

$$\mathbf{J}_L = -\mathbf{J}_F, \qquad \rho_L = -\rho_F. \tag{1.4}$$

Substitution of $\mathbf{E}_L = -\mathbf{E}_M$ therefore gives

$$\begin{aligned} \boldsymbol{\nabla} \times \mathbf{E}_L &= +\dot{\mathbf{B}}, & \boldsymbol{\nabla} \cdot \mathbf{B} &= 0, \\ \boldsymbol{\nabla} \times \mathbf{B} &= -\dot{\mathbf{E}}_L - \mathbf{J}_L, & \boldsymbol{\nabla} \cdot \mathbf{E}_L &= \rho_L. \end{aligned} \tag{1.5}$$

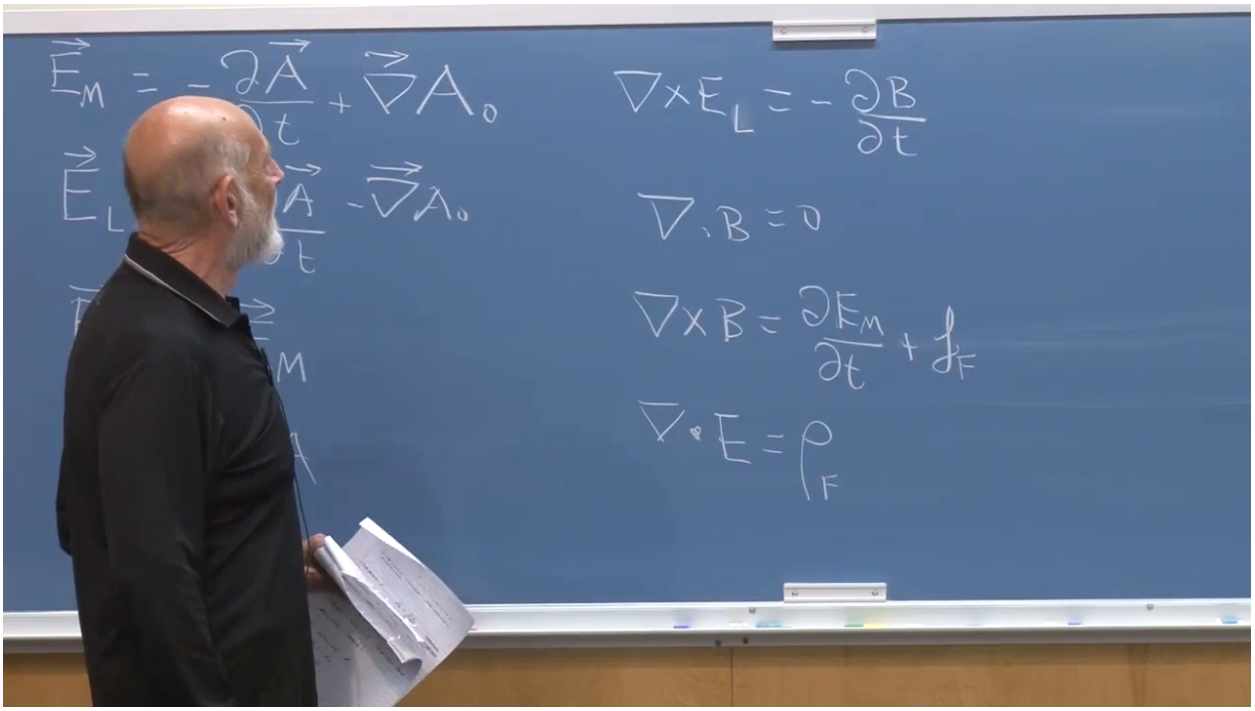

Figure 1.1: The complete board comparison between the Maxwell–Franklin and course conventions. *Lecture frame: 00:07:10.*

Nothing measurable depends on which dictionary is chosen. What matters is consistency, especially when a calculated field is compared with a published experimental value. The practical joke is that such direct numerical comparisons had arisen so rarely in the lecturer's own work that the opposite sign survived unnoticed until the class called attention to it. For the electromagnetic calculation later in this chapter we shall use the standard field $\mathbf{E} \equiv \mathbf{E}_M$.

1.2 Three Different Quantities Called Momentum

The word *momentum* now has to be unpacked. Mechanical momentum, canonical momentum, and Noether momentum can coincide in simple problems, but they are not three descriptions of one quantity. In a general system they can have different numerical values.

1.2.1 Mechanical momentum

For a body of total mass M, mechanical momentum is

$$\mathbf{P}_{\text{mech}} = M\dot{\mathbf{X}}_{\text{cm}}. \tag{1.6}$$

For one particle, $\mathbf{X}_{\mathrm{cm}}$ is simply its position. For a composite body, the center-of-mass qualification is essential.

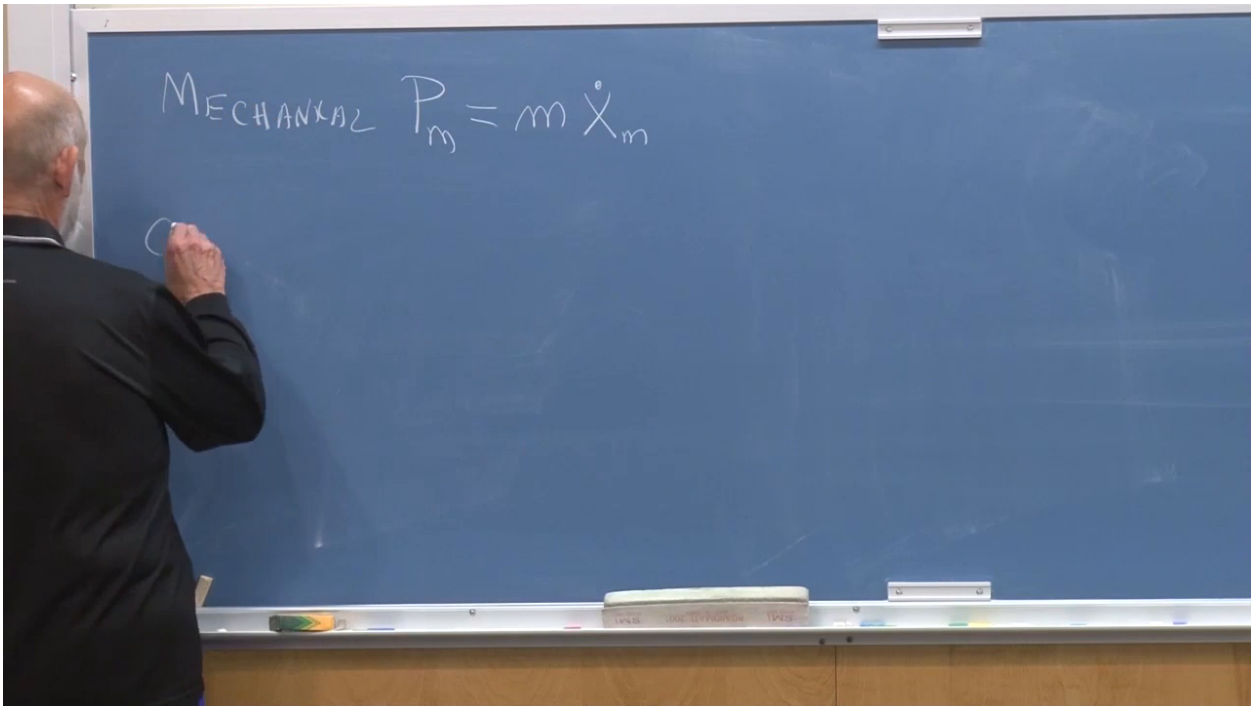

Figure 1.2: Mechanical momentum introduced as mass times the center-of-mass velocity. *Lecture frame: 00:11:15.*

1.2.2 Canonical momentum

Suppose the mechanics is described by $L(q_i, \dot{q}_i)$. The canonical momentum conjugate to q_i is

$$p_i = \frac{\partial L}{\partial \dot{q}_i}. \tag{1.7}$$

For

$$L = \frac{1}{2} m \dot{\mathbf{x}}^2 - V(\mathbf{x}), \tag{1.8}$$

this gives $\mathbf{p} = m\dot{\mathbf{x}}$, so canonical and mechanical momentum happen to agree.

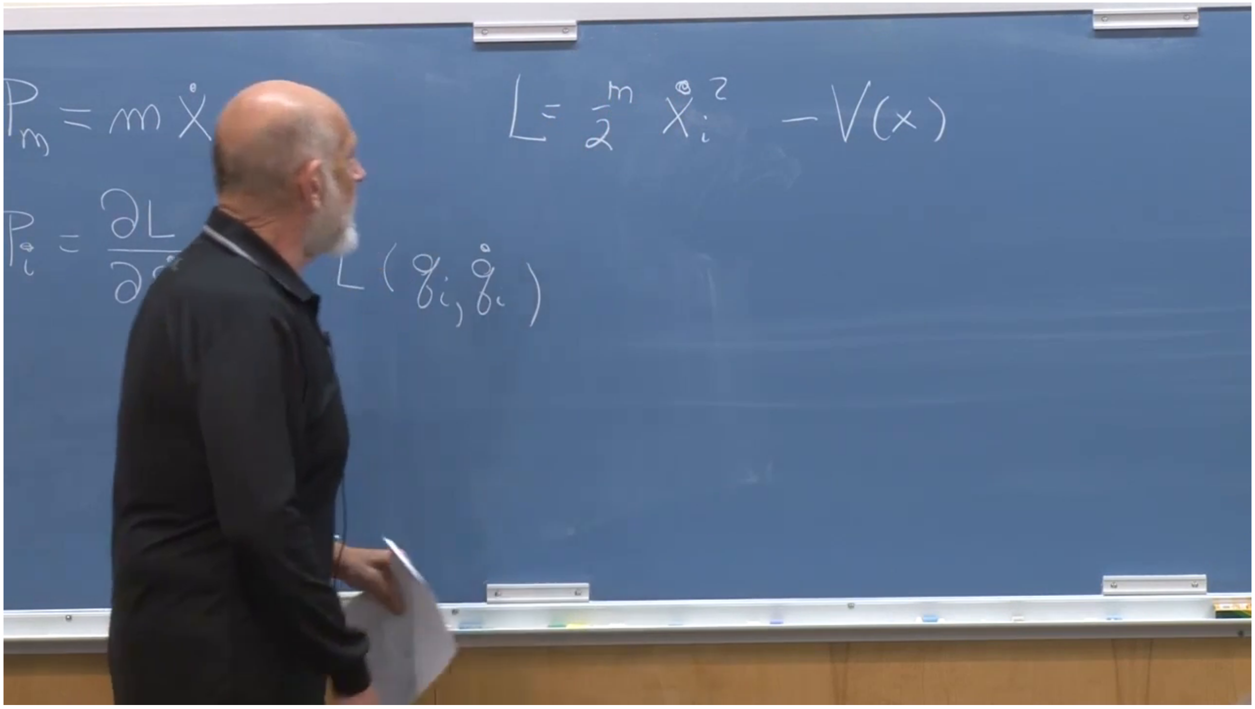

Figure 1.3: Mechanical and canonical momentum displayed beside the simple particle Lagrangian. *Lecture frame: 00:13:40.*

A charged particle in a magnetic field is the useful counterexample. With the sign convention written on the board,

$$L = \frac{1}{2}m\dot{\mathbf{x}}^2 - V(\mathbf{x}) - e\,\dot{\mathbf{x}} \cdot \mathbf{A}(\mathbf{x}), \tag{1.9}$$

and therefore

$$p_i = m\dot{x}_i - eA_i. \tag{1.10}$$

Changing the convention for charge or for the interaction term changes the displayed sign, but not the lesson: the canonical momentum contains a vector-potential contribution and is not merely $m\dot{x}_i$. Hamilton's equations use the canonical variable. One may later move the field-dependent term to the other side of an equation and recover a Newtonian-looking acceleration, but that rearrangement does not change the definition.

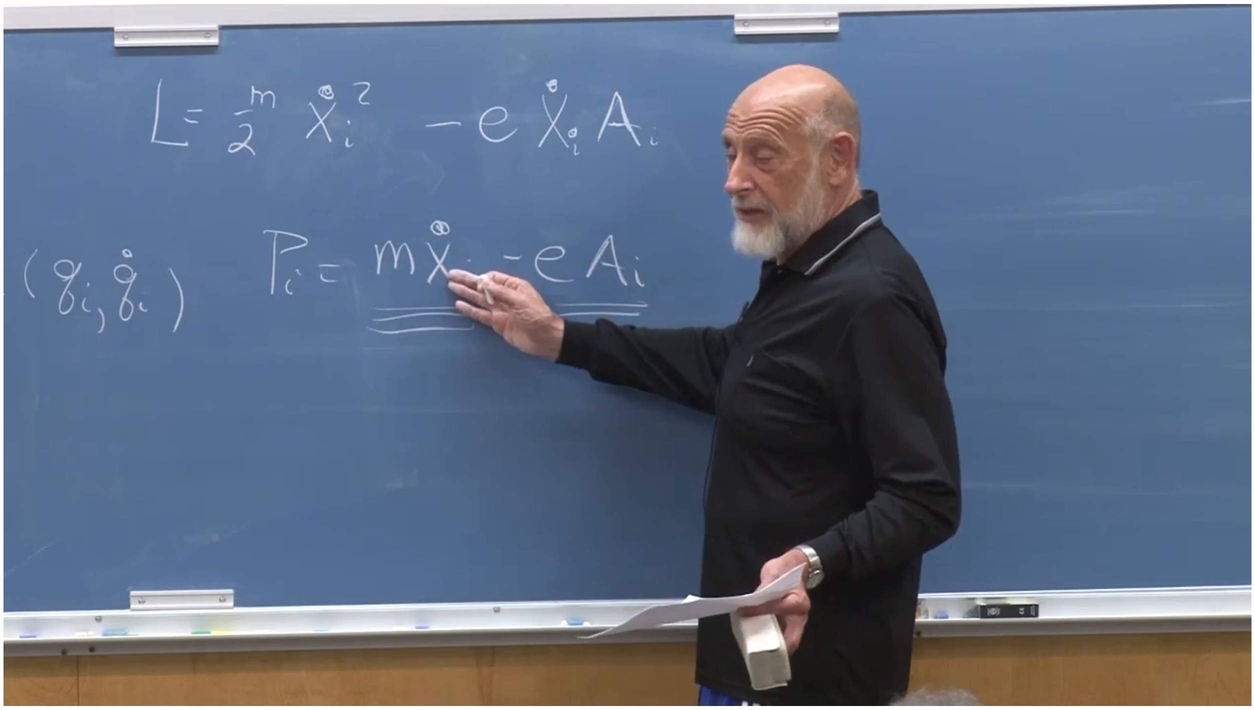

Figure 1.4: The vector-potential term shifts canonical momentum away from mechanical momentum. *Lecture frame: 00:15:50.*

1.2.3 Noether momentum

The third momentum is the conserved charge associated with spatial translation invariance. For an infinitesimal symmetry

$$q_i \longrightarrow q_i + \delta q_i \tag{1.11}$$

that leaves the Lagrangian invariant, Noether's theorem gives, up to the normalization of the infinitesimal parameter,

$$Q = \sum_i p_i \, \delta q_i. \tag{1.12}$$

The p_i appearing here are canonical momenta. If the symmetry is a uniform translation and canonical momentum happens to equal mechanical momentum, Q reduces to the familiar total momentum. In a less simple system that coincidence must be proved, not assumed.

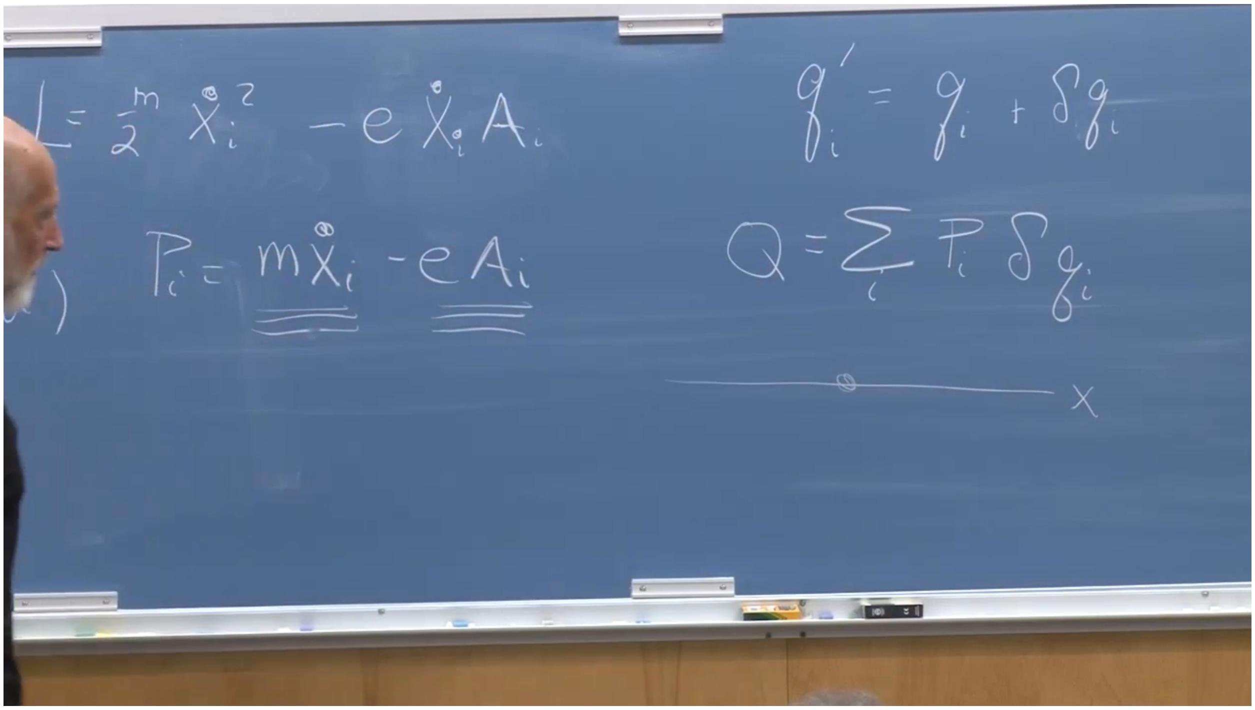

Figure 1.5: The Noether charge assembled from canonical momenta and the infinitesimal changes of the coordinates.

Lecture frame: 00:19:00.

1.3 Energy as the Hamiltonian

Spatial translations lead to momentum; time translations lead to energy. If the Lagrangian has no explicit time dependence, the Hamiltonian

$$H = \sum_i p_i \dot{q}_i - L \tag{1.13}$$

is conserved. This is the Legendre transform of the Lagrangian, not a rule about changing signs by inspection.

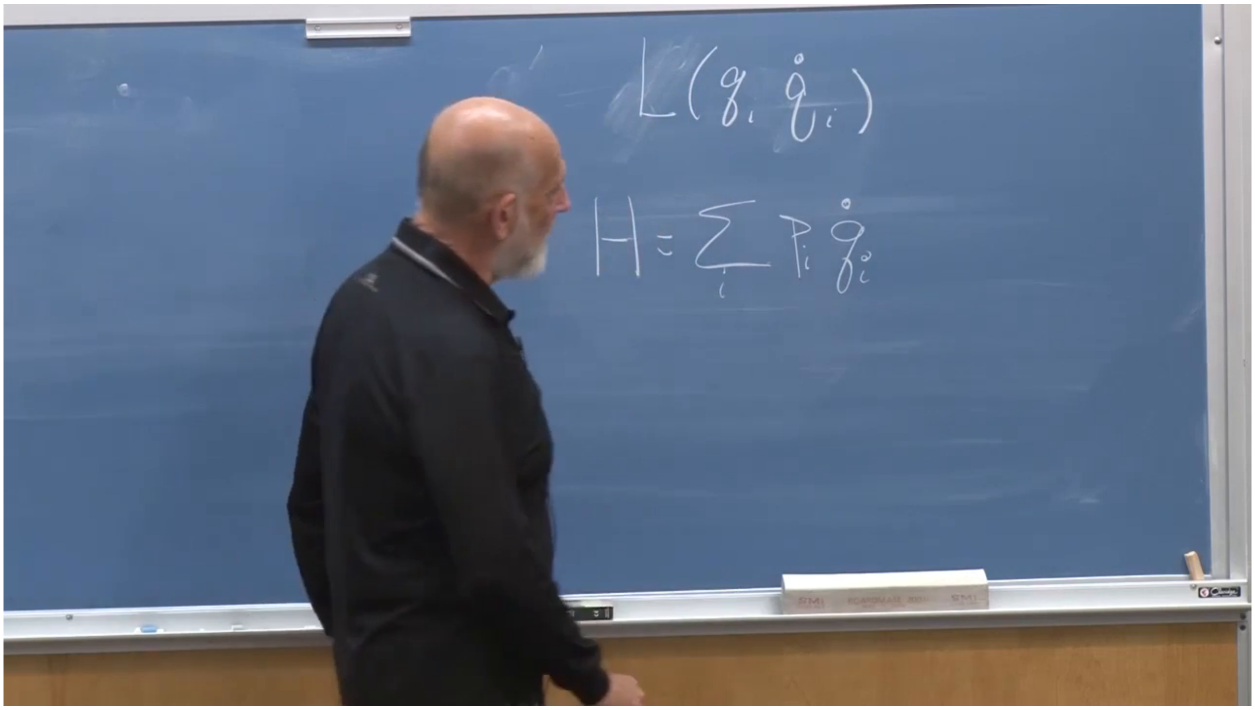

Figure 1.6: The Hamiltonian written as the Legendre transform of a general Lagrangian.
Lecture frame: 00:22:55.

For the one-particle example,

$$\begin{aligned} p &= m\dot{x}, \\ H &= p\dot{x} - L = m\dot{x}^2 - \left(\frac{1}{2}m\dot{x}^2 - V(x)\right) = \frac{1}{2}m\dot{x}^2 + V(x). \end{aligned} \tag{1.14}$$

Thus the Lagrangian is kinetic minus potential energy, whereas the Hamiltonian is kinetic plus potential energy.

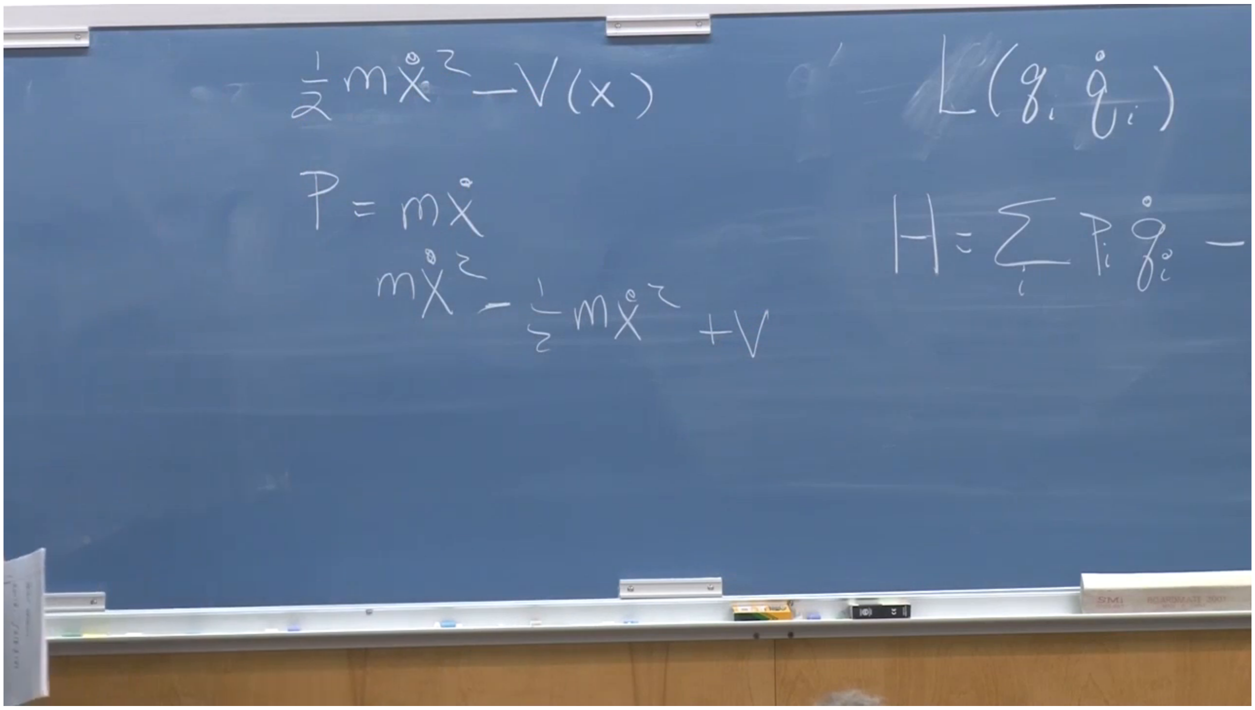

Figure 1.7: The particle calculation showing how $H = T + V$ follows from $H = p\dot{x} - L$. *Lecture frame: 00:24:20.*

When every velocity dependence is a conventional positive quadratic kinetic term and the remainder contains no velocities, one can read off the Hamiltonian by reversing the sign of the nonvelocity part. That shortcut will be used repeatedly. It is safe only because the Legendre transform has this simple form in the examples at hand.

1.4 A Field Is a Mechanical System with Continuously Many Coordinates

To expose the mechanical structure of field theory, choose one inertial frame and temporarily stop insisting that every line display manifest Lorentz covariance. Classical mechanics has a time axis and coordinates $q_i(t)$. A scalar field has a time axis and coordinates $\phi(\mathbf{x}, t)$. The discrete label i has simply become the continuous label $\mathbf{x}$:

$$q_i(t) \quad \longrightarrow \quad \phi(\mathbf{x}, t). \tag{1.15}$$

The value of the field at each spatial point is one degree of freedom. Here $\mathbf{x}$ labels which degree of freedom is meant; it is not itself the generalized coordinate.

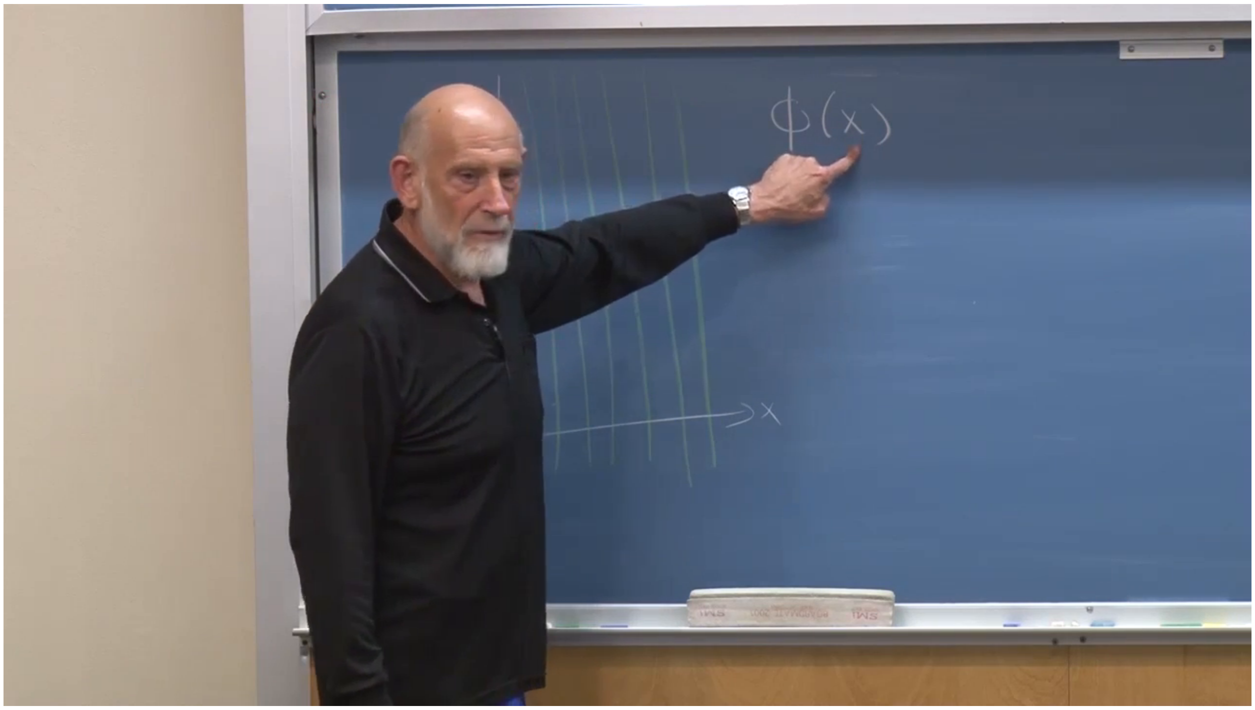

Figure 1.8: A spatial lattice used to display the field value at each point as a separate mechanical degree of freedom.

Lecture frame: 00:30:15.

Question from the audience. Why not write the field as $\phi(\mathbf{x}, t)$ from the beginning?

Response. That is the complete notation. The temporary $\phi(\mathbf{x})$ suppresses the common time argument just as one often writes q_i while remembering that every q_i depends on time. The label i, or now $\mathbf{x}$, tells us which degree of freedom is under discussion. [a]

[a] Lecture timestamp: 00:32:43.

Question from the audience. Are the field values at neighboring points really independent coordinates? What imposes continuity?

Response. They are independent canonical coordinates. Smoothness is not imposed by identifying neighboring values; it is selected dynamically. The Hamiltonian contains a positive gradient-energy term, so a fixed jump over an ever smaller distance costs ever more energy. Finite-energy configurations are consequently smooth in the continuum limit. [a]

[a] Lecture timestamp: 00:33:09.

A spatial derivative makes the coupling between neighboring degrees of freedom explicit. On a lattice of spacing ϵ,

$$\partial_x \phi(x,t) = \lim_{\epsilon \to 0} \frac{\phi(x+\epsilon, t) - \phi(x,t)}{\epsilon}. \tag{1.16}$$

It depends on two nearby coordinates, but it contains no generalized velocity. In the mechanical analogy it belongs to the coordinate dependence of the Lagrangian, not to the $\dot{q}_i$ dependence.

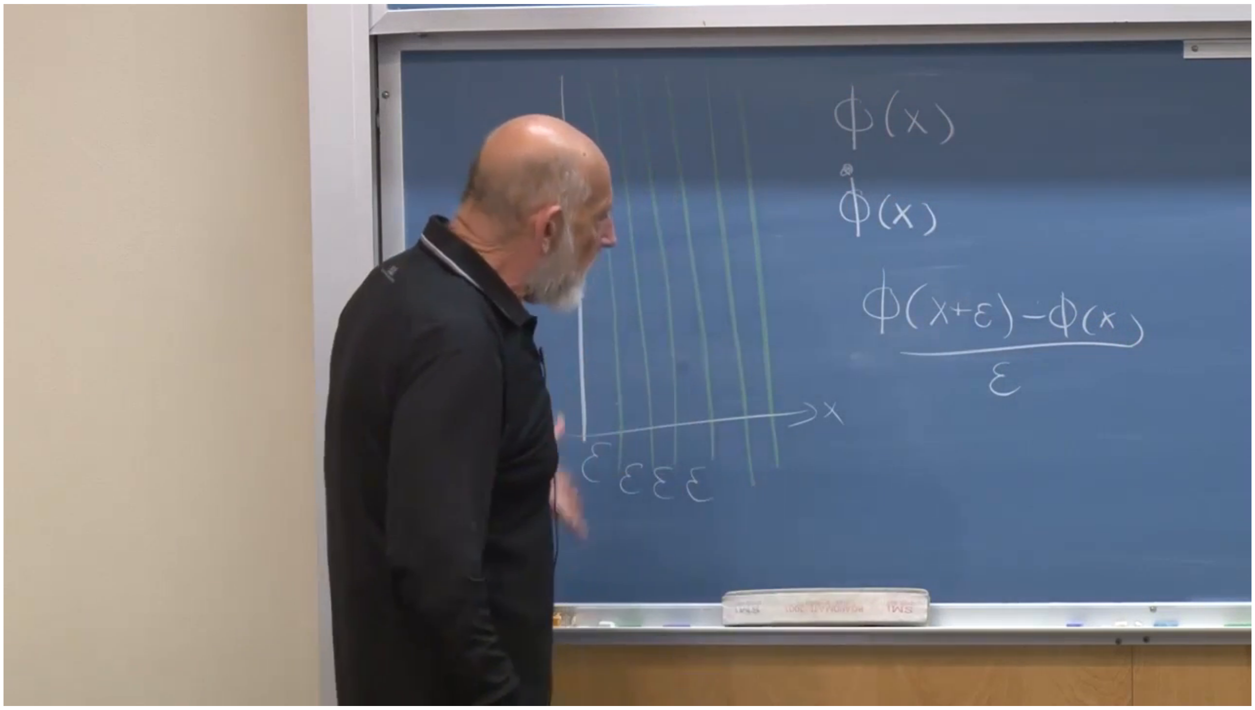

Figure 1.9: The spatial derivative reconstructed as a neighboring-field difference on the lattice. *Lecture frame: 00:31:40.*

The action can now be written in explicitly mechanical form:

$$
\begin{aligned}
S &= \int dt\, L, \\
L &= \int d^3x\, \mathcal{L}\Big(\phi, \dot{\phi}, \boldsymbol{\nabla}\phi\Big)\,, \\
S &= \int dt\, d^3x\, \mathcal{L}\Big(\phi, \dot{\phi}, \boldsymbol{\nabla}\phi\Big)\,.
\end{aligned}
\tag{1.17}
$$

The script letter $\mathcal{L}$ is the Lagrangian density; integrating it over space gives the ordinary Lagrangian that is integrated over time. A local theory couples each point only through fields and their derivatives at that point. Splitting time from space hides Lorentz covariance for a moment, but makes the mechanics transparent.

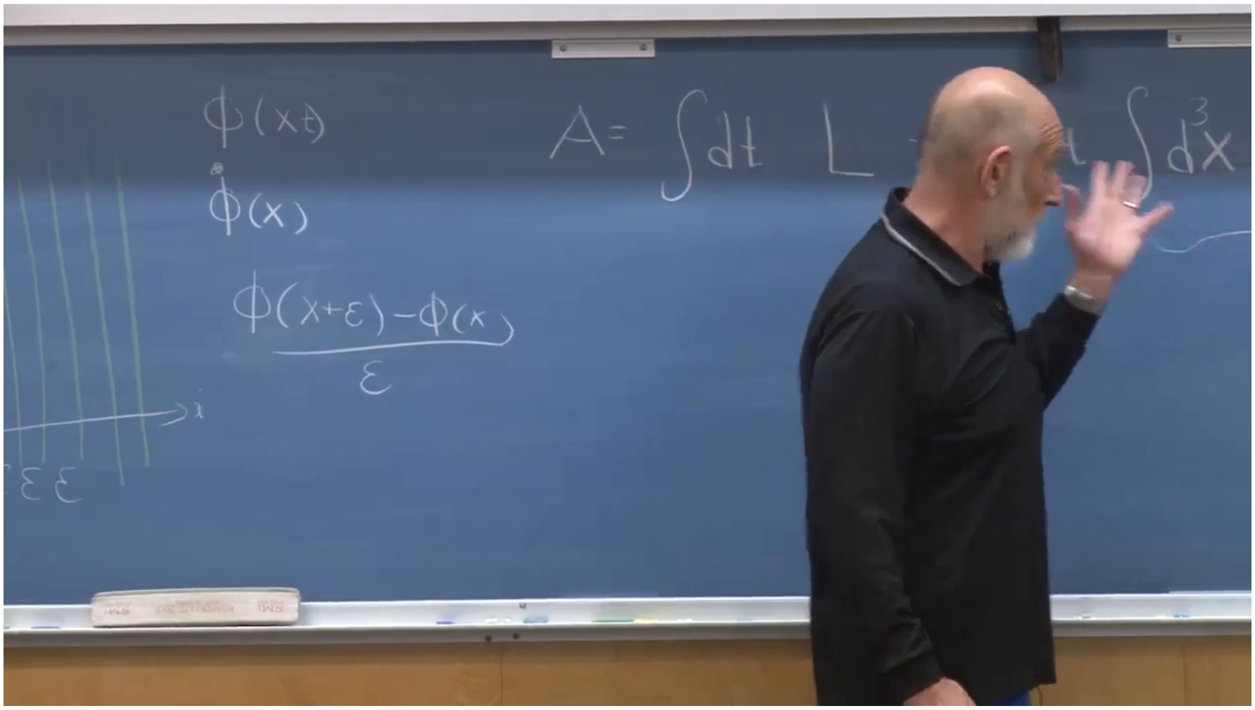

Figure 1.10: The field action separated into a time integral and a spatial integral of the Lagrangian density.

Lecture frame: 00:36:40.

1.5 Canonical Field Momentum and Hamiltonian Density

The velocity of a field coordinate is $\dot{\phi}(\mathbf{x}, t)$: the rate at which the field changes in time at one fixed point. It is not motion from one spatial point to another. The canonical momentum conjugate to the field is therefore itself a field,

$$\Pi(\mathbf{x}, t) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}(\mathbf{x}, t)}. \tag{1.18}$$

For several fields ϕ_a, there is one momentum field Π_a for each.

The discrete sum in the Legendre transform becomes a spatial integral:

$$\begin{aligned} H &= \int d^3x \left[\Pi\dot{\phi} - \mathcal{L}\right] = \int d^3x\, \mathcal{H}, \\ \mathcal{H} &= \Pi\dot{\phi} - \mathcal{L}. \end{aligned} \tag{1.19}$$

The quantity inside the integral is the energy density.

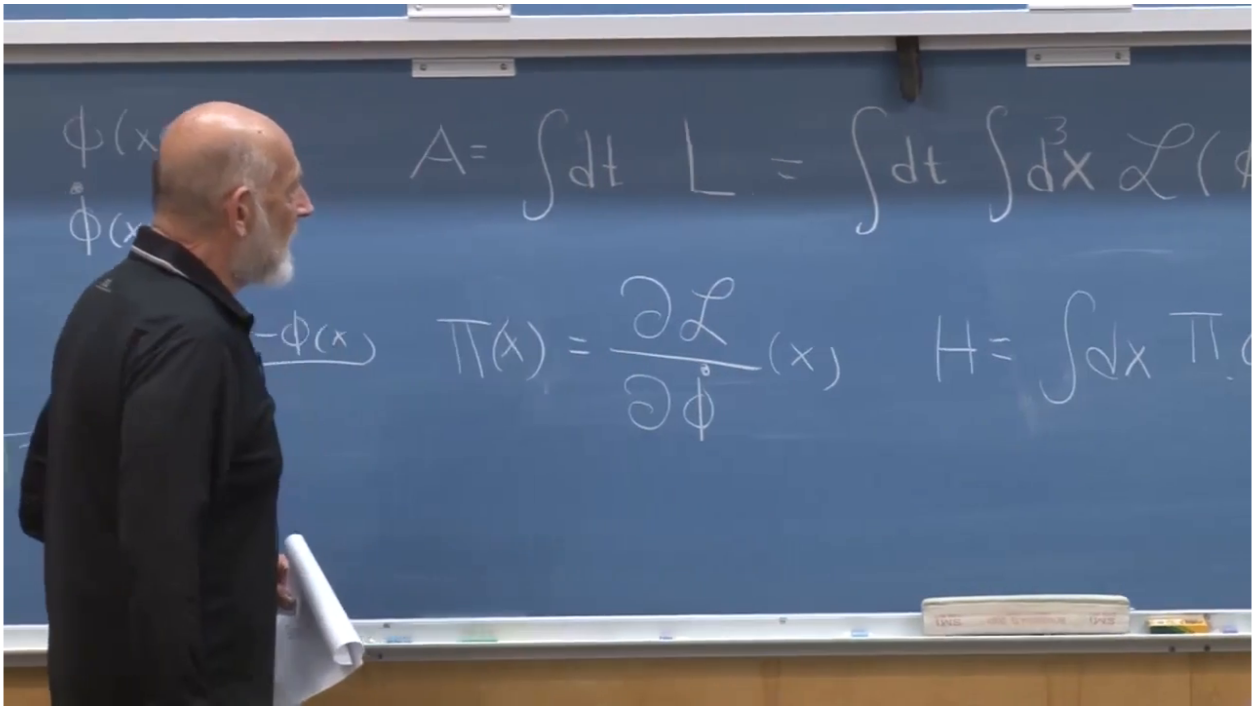

Figure 1.11: The field Legendre transform and its interpretation as an integral of local energy density.

Lecture frame: 00:42:40.

Question from the audience. Why may the derivative of the total Lagrangian with respect to $\dot{\phi}$ at one point be replaced by a derivative of the local density there?

Response. Think of the spatial integral as a continuum sum. Under the locality assumption used here, the velocity $\dot{\phi}(\mathbf{x})$ appears only in the term belonging to that same point. Differentiation with respect to that particular velocity removes all the other terms. Spatial derivatives may couple neighboring field values, but they do not introduce neighboring time derivatives in this theory. [a]

[a] Lecture timestamp: 00:51:56.

1.5.1 The scalar field

In one spatial dimension, take

$$\mathcal{L} = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\partial_x \phi)^2 - V(\phi). \tag{1.20}$$

This is the noncovariant display of the Lorentz scalar used earlier: the relative minus sign between temporal and spatial derivatives comes from the Minkowski metric. The canonical momentum is

$$\Pi = \dot{\phi}, \tag{1.21}$$

and the Hamiltonian becomes

$$H = \int dx \left[\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\partial_x \phi)^2 + V(\phi)\right]. \tag{1.22}$$

During the calculation the potential was momentarily called $V(x)$; the class corrected it to $V(\phi)$. It may depend on position indirectly through the field, but in the translation-invariant theory it has no prescribed external x-dependence.

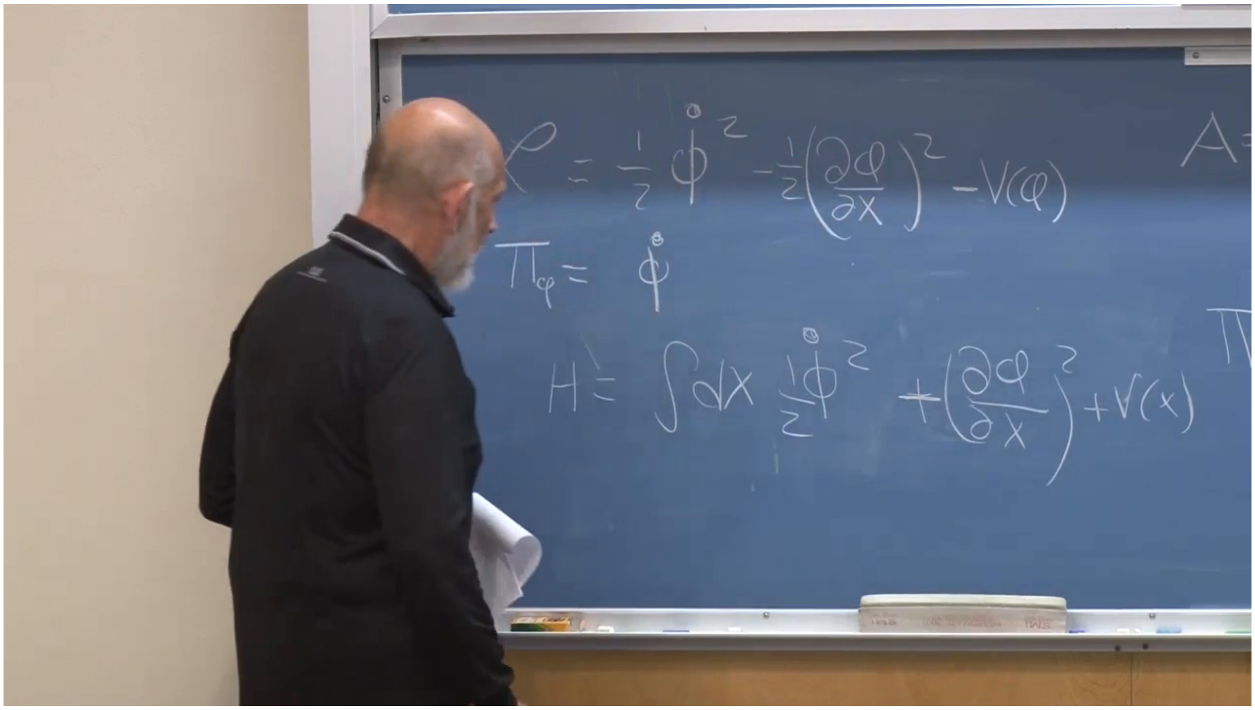

Figure 1.12: Scalar Lagrangian density, conjugate momentum, and positive Hamiltonian assembled on one board.

Lecture frame: 00:47:05.

The whole part without time derivatives plays the role of potential energy. That includes both $V(\phi)$ and the gradient energy $\frac{1}{2}(\partial_x\phi)^2$. The Lagrangian can have either sign; a stable energy should be bounded below.

Question from the audience. May the potential $V(\phi)$ be negative?

Response. Yes. If it has a finite lower bound, a constant may be added so that its minimum is zero without changing the classical field equation. A potential unbounded below describes an instability: the field can lower its energy without limit. The stable vacuum is a constant field sitting at a minimum of the shifted potential. [a]

[a]Lecture timestamp: 00:49:03.

The earlier continuity question is now answered quantitatively. A jump $\Delta\phi$ across a lattice spacing ϵ contributes

$$\frac{1}{2}(\partial_x\phi)^2 \sim \frac{1}{2}\left(\frac{\Delta\phi}{\epsilon}\right)^2. \tag{1.23}$$

As $\epsilon \to 0$, a fixed jump gives an arbitrarily large energy density. Finite energy therefore suppresses jagged configurations.

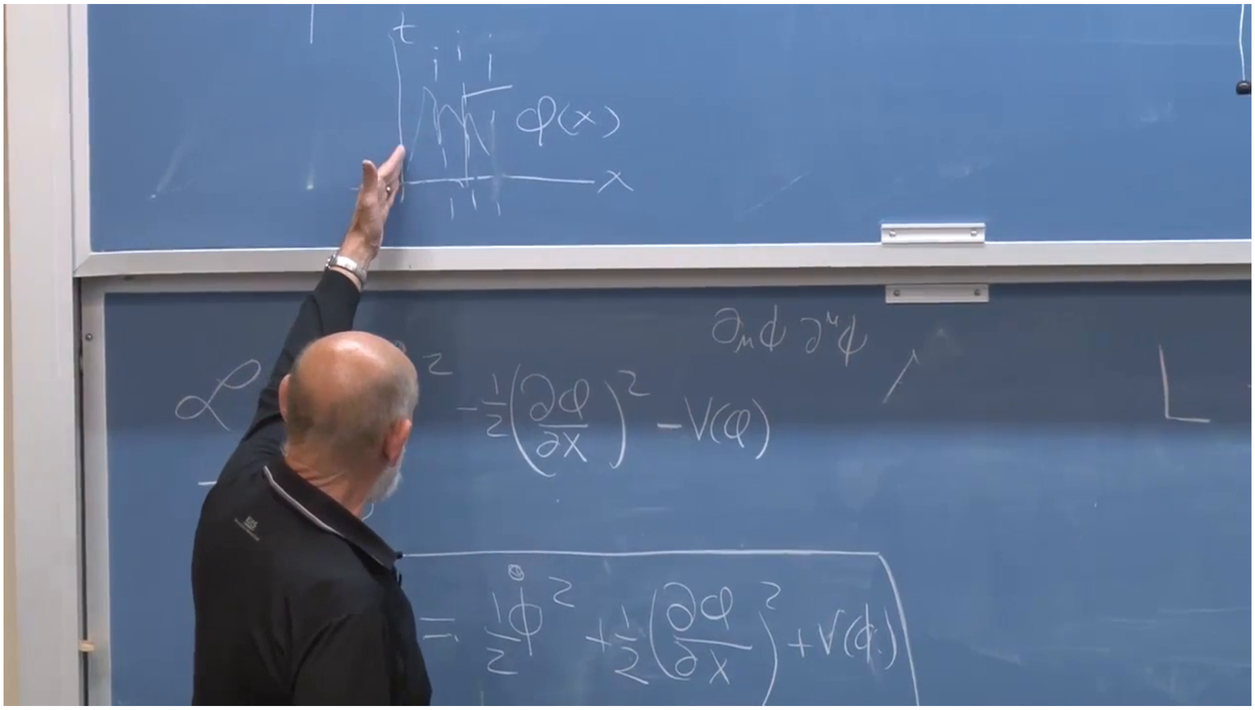

Figure 1.13: The scalar energy formula retained while the board sketch contrasts a smooth field with a sharp variation.

Lecture frame: 00:54:20.

1.6 Local Energy and the Meaning of T^{00}

The energy density receives a compact name:

$$T^{00} = \mathcal{H} = \Pi\dot{\phi} - \mathcal{L}. \tag{1.24}$$

The two zero indices encode two separate statements. Energy is the time component P^0 of the energy–momentum four-vector, while a density is the time component of a conserved current. Thus T^{00} is the density of the time component of four-momentum.

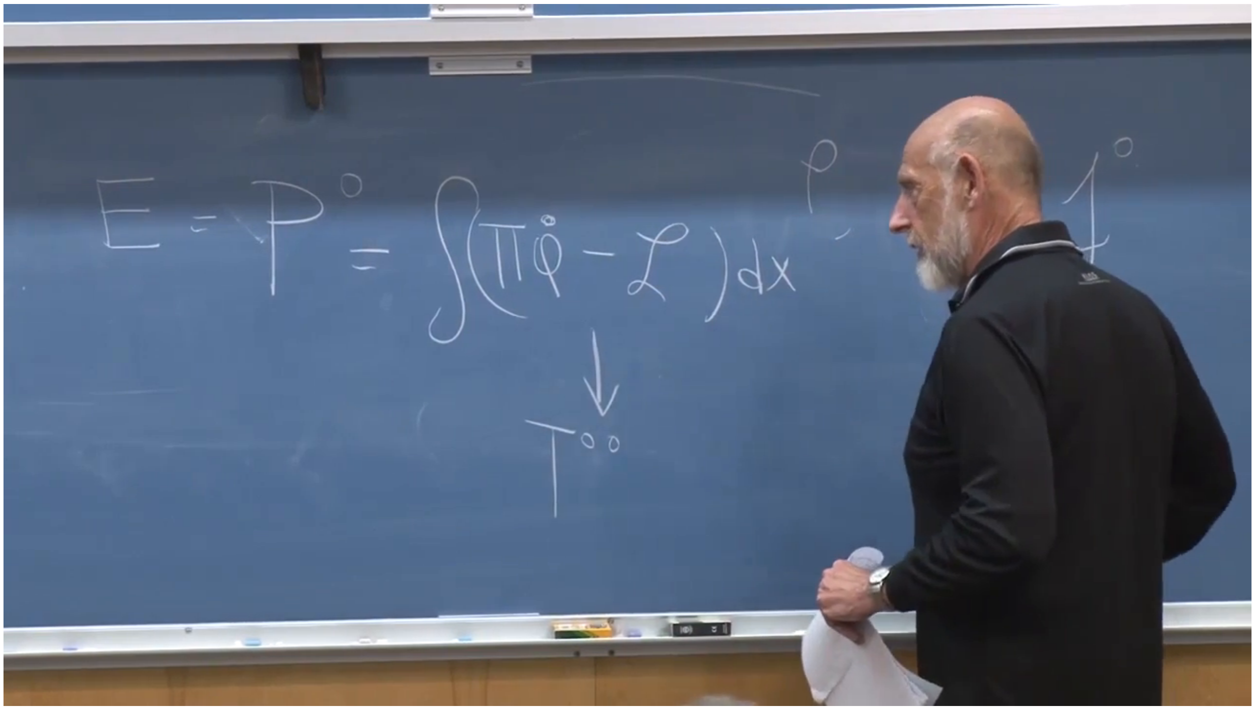

Figure 1.14: The field energy written locally and identified with T^{00}.
Lecture frame: 01:00:00.

Conservation in a relativistic local theory is not merely a statement that one global number remains unchanged. Energy cannot disappear in the laboratory and reappear at Alpha Centauri. A change inside a region must be explained by a flux through its boundary. In local form,

$$\partial_\mu T^{\mu\nu} = 0, \tag{1.25}$$

so the first index labels density or flux direction, and the second labels which component P^ν is being transported.

Question from the audience. Does calling T^{00} a density mean that the total energy is obtained by integrating it?

Response. Exactly: $E = P^0 = \int d^3x\, T^{00}$. The local density is the integrand in the field Legendre transform. Its variation at one place is balanced by an energy current through neighboring places. [a]

[a] Lecture timestamp: 00:59:06.

1.7 Momentum Density from Spatial Translations

Return to Noether's formula, now replacing the discrete sum by a spatial integral. Under a small translation in the m-direction, the field changes by

$$\delta\phi = \phi(\mathbf{x}) - \phi(\mathbf{x} - \boldsymbol{\epsilon}) = \epsilon^m \partial_m \phi + O(\epsilon^2), \tag{1.26}$$

for the passive convention used on the board. Reversing active and passive translations reverses the overall sign; the physical generator is fixed once that convention is chosen. Factoring out ϵ^m, the Noether momentum has the structure

$$P_m \;\propto\; \int d^3x\, \Pi\, \partial_m \phi. \tag{1.27}$$

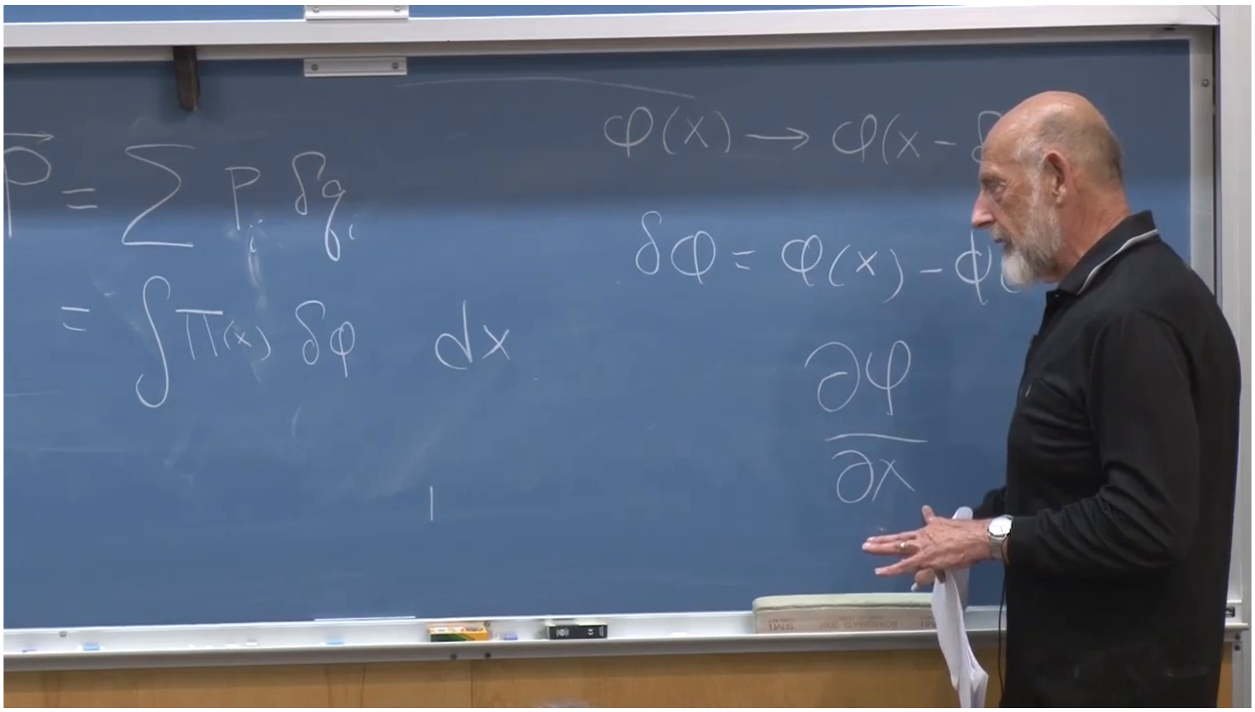

Figure 1.15: A translated field profile and the derivative that measures its infinitesimal change. *Lecture frame: 01:04:20.*

Question from the audience. Why was the infinitesimal translation ϵ not kept in the momentum formula?

Response. The Noether charge is first obtained proportional to ϵ. Momentum is defined as the coefficient of that arbitrary translation parameter, so ϵ is divided out. Multiplying a conserved quantity by a fixed constant would not affect conservation, but extracting the coefficient gives the standard normalization. [a]

[a] Lecture timestamp: 01:04:45.

Question from the audience. Does the translation interval become the dx in the continuum integral?

Response. They arise from different limits. The measure dx comes from replacing the sum over spatially labeled degrees of freedom by an integral. The translation parameter ϵ describes the symmetry operation and is factored out of its Noether charge. Both are infinitesimal in intermediate reasoning, but they play different roles. [a]

[a] Lecture timestamp: 01:05:16.

In the notation of the lecture, the density of the m-component of momentum is

$$T^{0m} = \Pi\,\partial_m\phi, \tag{1.28}$$

with the understood translation-sign convention. For the scalar theory $\Pi = \dot{\phi}$, so it is proportional to $\dot{\phi}\,\partial_m\phi$.

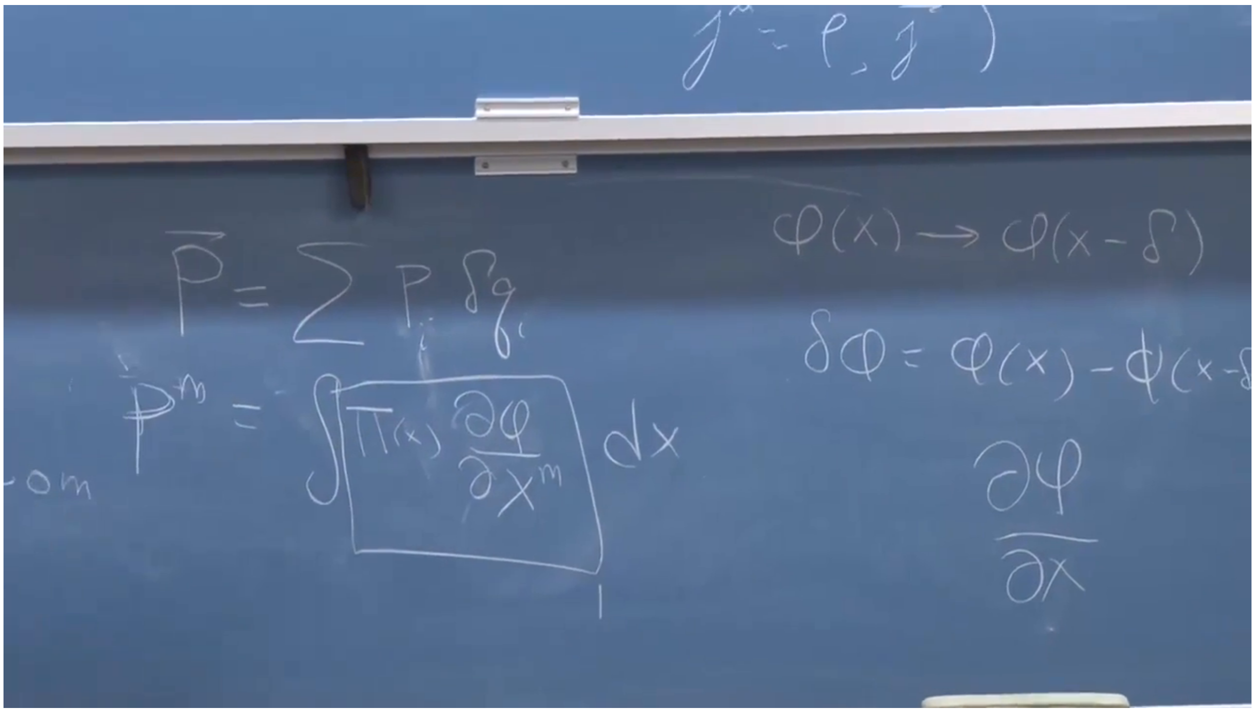

Figure 1.16: The Noether momentum density $T^{0m} = \Pi\,\partial_m\phi$ on the board.
Lecture frame: 01:07:10.

This is literal momentum carried by a wave. Translation invariance applies only when everything is translated: fields, particles, sources, and apparatus. A light pulse absorbed by a freely suspended board gives the board a recoil. Ordinary light makes the effect tiny; a sufficiently powerful laser can transfer enough momentum to compress a small target. The conserved total belongs to particles and fields together.

1.7.1 What replaces Newton's equation?

The field Euler–Lagrange equation derived from Eq. (1.20) is

$$\ddot{\phi} - \nabla^2\phi = -\frac{dV}{d\phi}. \tag{1.29}$$

It is the field-theory analogue of Newton's law.

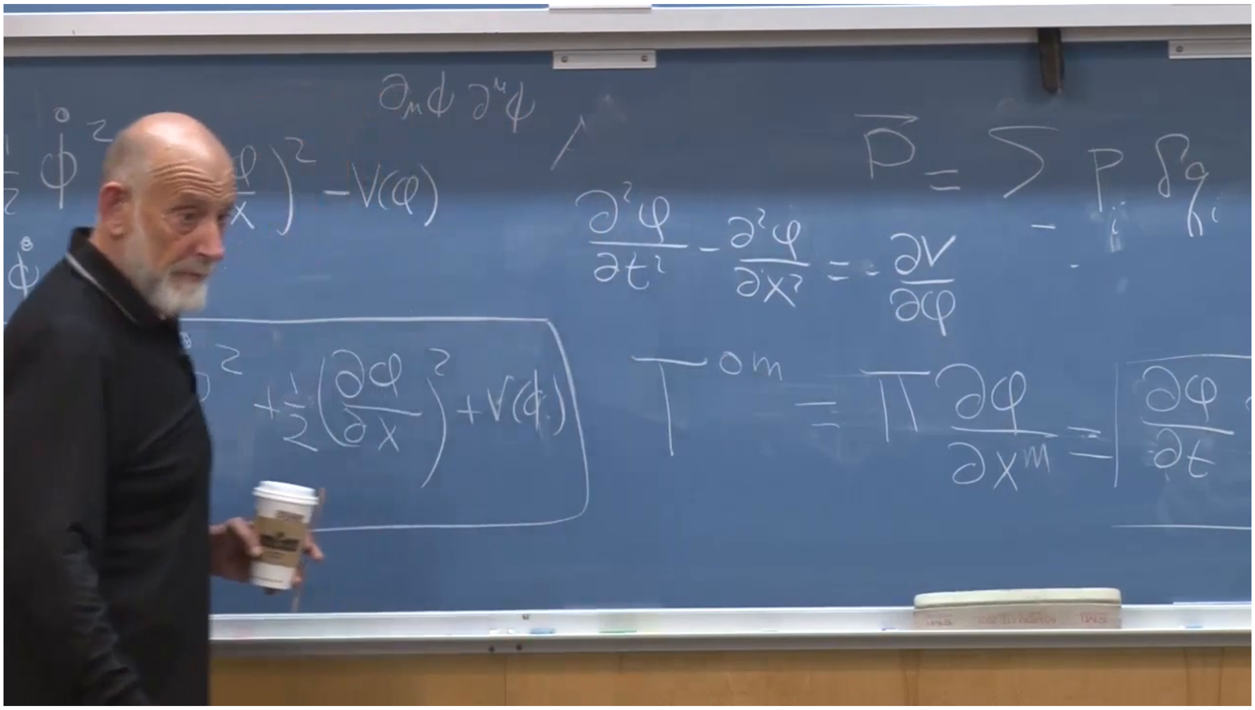

Figure 1.17: The scalar field equation displayed beside its Hamiltonian and momentum density. *Lecture frame: 01:12:55.*

Question from the audience. What corresponds to Newton's law for a field?

Response. The Euler–Lagrange field equation does. Here it is a wave equation with the force term $-V'(\phi)$. The role of $m\ddot{x} = -V'(x)$ is played by $\ddot{\phi} - \nabla^2\phi = -V'(\phi)$. [a]

[a] Lecture timestamp: 01:11:48.

Question from the audience. How do we know that a proposed field Lagrangian is correct? Must we already know the field equation?

Response. Either direction is possible. Historically one may infer equations from experiment and then seek an action, as happened with Newton and Maxwell. In modern theory one often starts from the field content, locality, symmetry, and a small set of admissible terms, derives the equations, and tests their predictions. The displayed scalar could describe a fundamental field, but it could equally describe the transverse displacement of a rope whose continuum action is derived from its constituent particles. [a]

[a] Lecture timestamp: 01:12:55.

1.8 The Electromagnetic Field in Temporal Gauge

The components already found, T^{00} and T^{0m}, belong to the energy–momentum tensor. Its remaining components describe energy and momentum flux. This tensor will later source gravity, but electromagnetism provides the immediate example.

The basic electromagnetic coordinate is the four-potential A_μ. Gauge freedom allows

$$A_\mu \longrightarrow A'_\mu = A_\mu + \partial_\mu S \tag{1.30}$$

without changing $F_{\mu\nu}$. Choose S to satisfy

$$\partial_t S = -A_0. \tag{1.31}$$

For example, one may integrate A_0 in time at each fixed spatial point. The transformed time component then vanishes:

$$A'_0 = A_0 + \partial_t S = 0. \tag{1.32}$$

This is temporal gauge. It removes A_0 from the list of dynamical coordinates, although Gauss's law remains as a constraint and a time-independent residual gauge freedom remains.

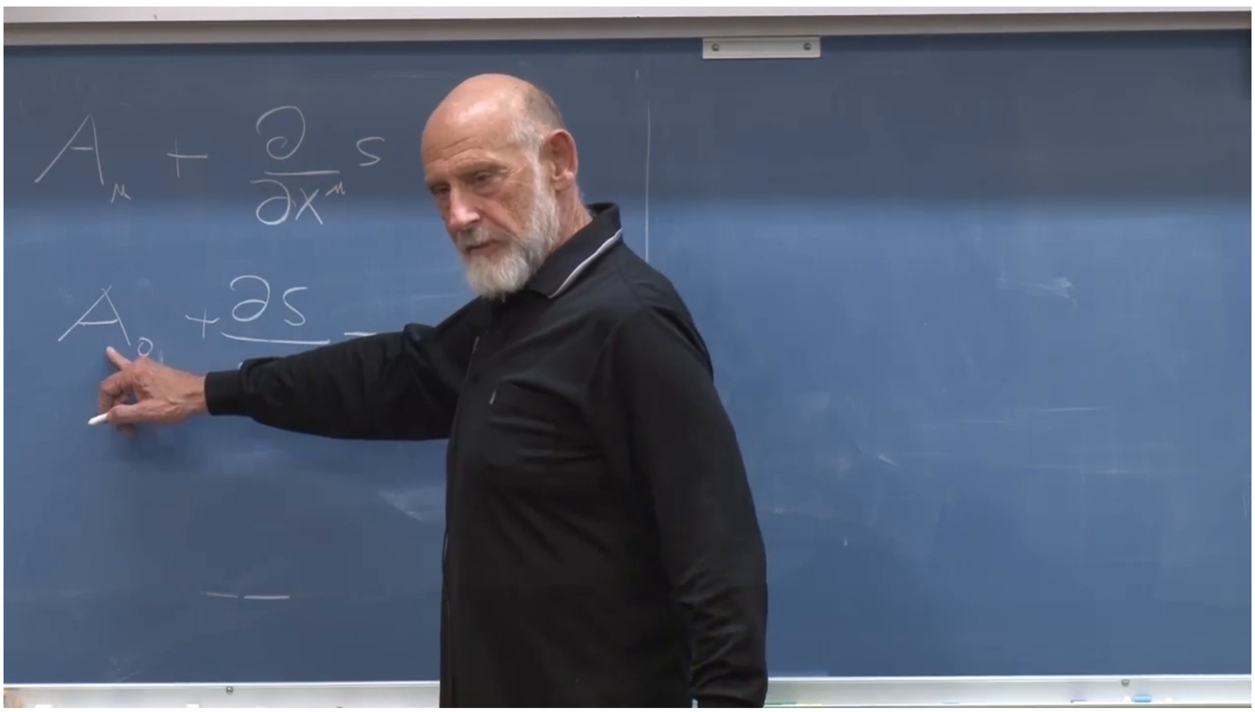

Figure 1.18: Gauge freedom used to choose $A_0 = 0$ by solving $\partial_t S = -A_0$.
Lecture frame: 01:20:15.

From this point onward let $\mathbf{E}$ denote the standard Maxwell field. Before gauge fixing,

$$E_m = -\partial_t A_m + \partial_m A_0, \qquad \mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}. \tag{1.33}$$

In temporal gauge these reduce to

$$\mathbf{E} = -\dot{\mathbf{A}}, \qquad \mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}. \tag{1.34}$$

Time derivatives of $\mathbf{A}$ produce the electric field; spatial derivatives produce the magnetic field.

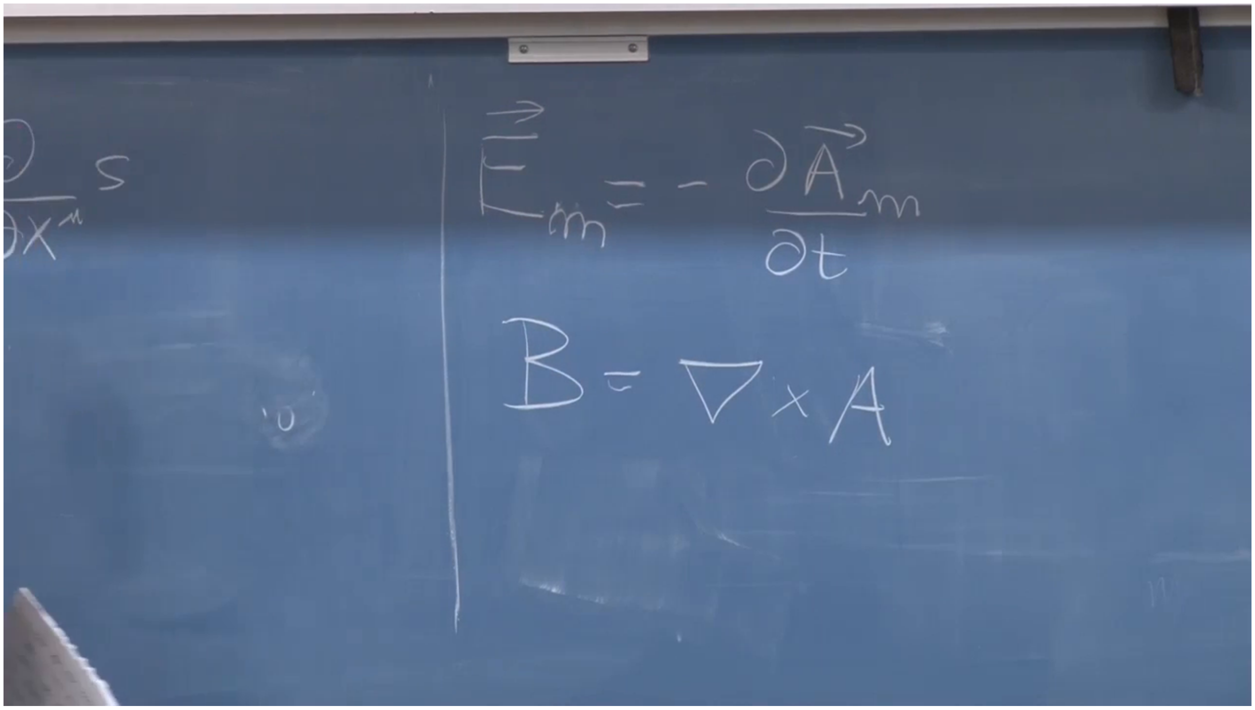

Figure 1.19: The electric and magnetic fields in temporal gauge: $\mathbf{E} = -\dot{\mathbf{A}}$ and $\mathbf{B} = \nabla \times \mathbf{A}$.
Lecture frame: 01:23:50.

1.9 Electromagnetic Canonical Momentum and Energy

The free Maxwell Lagrangian density is

$$\mathcal{L}_{\mathrm{EM}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = \frac{1}{2}\left(\mathbf{E}^2 - \mathbf{B}^2\right). \tag{1.35}$$

In temporal gauge,

$$\mathcal{L}_{\mathrm{EM}} = \frac{1}{2}\dot{\mathbf{A}}^{\,2} - \frac{1}{2}(\boldsymbol{\nabla} \times \mathbf{A})^2. \tag{1.36}$$

It has precisely the scalar-field pattern: squares of time derivatives minus squares of spatial derivatives.

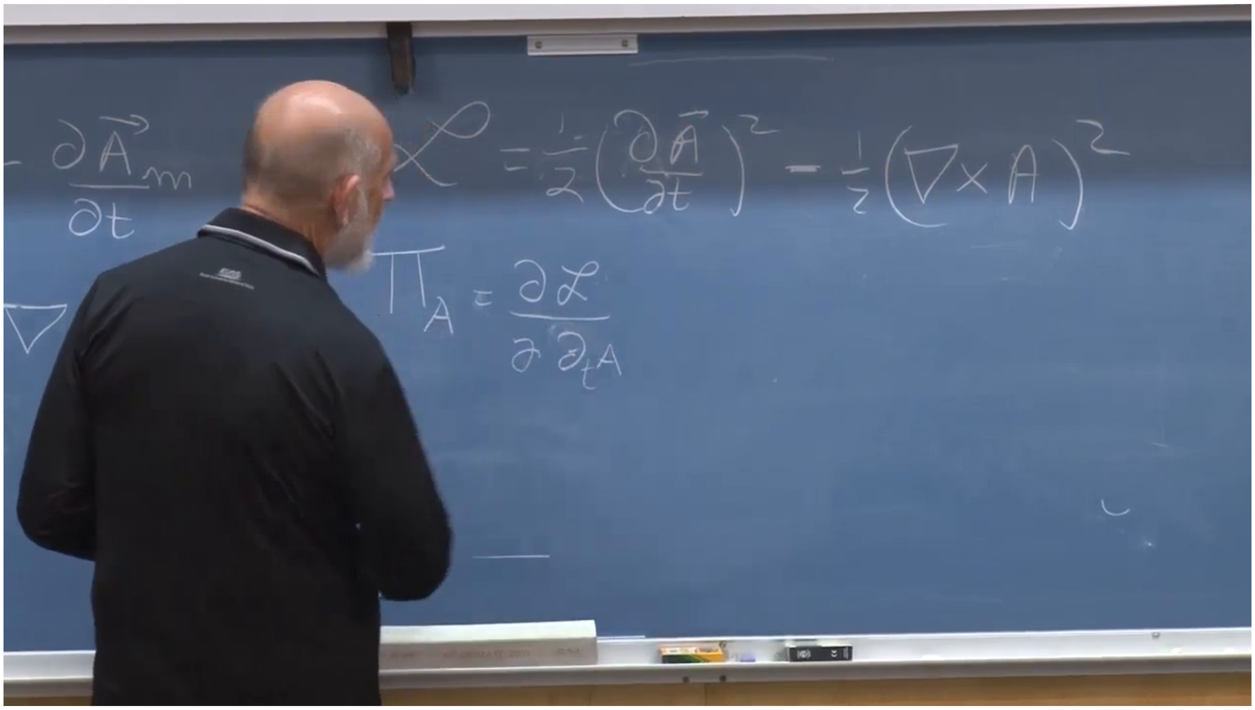

Figure 1.20: The Maxwell Lagrangian rewritten as a mechanical field Lagrangian for the spatial components of **A**.

Lecture frame: 01:28:10.

Each component A_m has a canonical momentum,

$$\Pi_m = \frac{\partial \mathcal{L}_{\text{EM}}}{\partial \dot{A}_m} = \dot{A}_m = -E_m. \tag{1.37}$$

This is a concrete example of why a convention matters. In the earlier opposite electric-field convention the final displayed sign would change, although the canonical derivative itself would not.

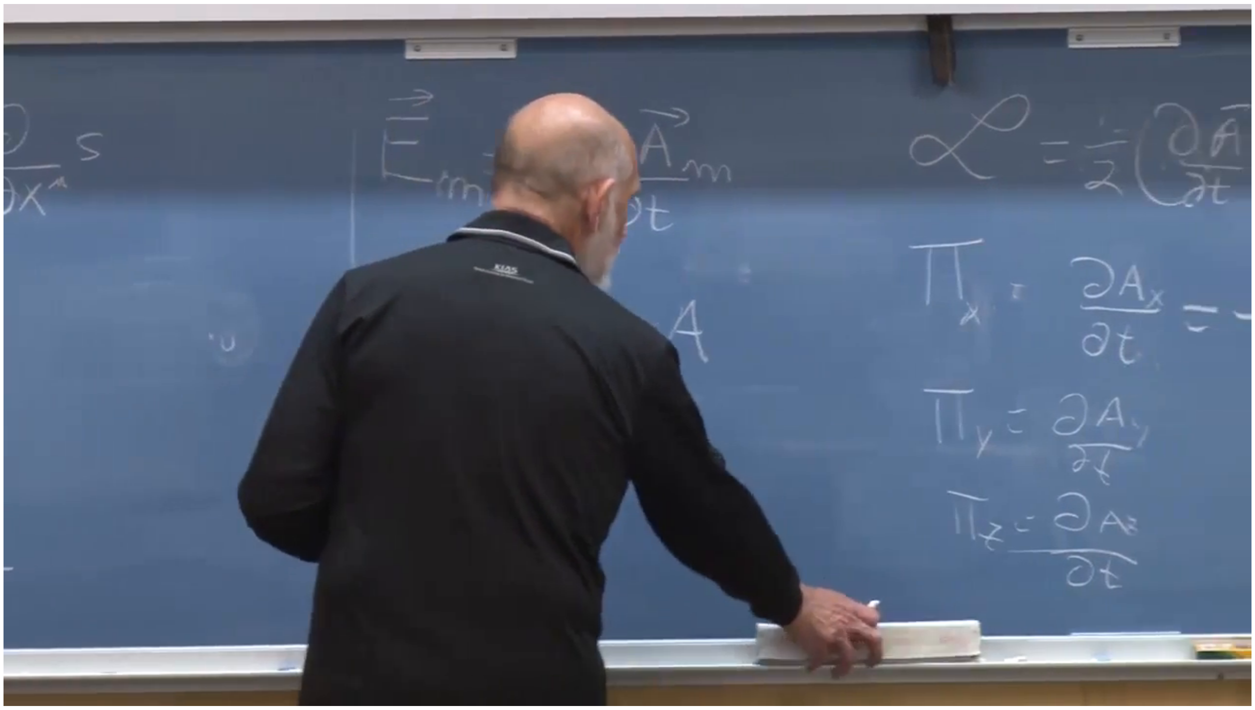

Figure 1.21: The three canonical momentum fields associated with A_x, A_y, A_z, each equal to the corresponding time derivative and therefore to minus the standard electric field.

Lecture frame: 01:30:20.

The Hamiltonian shortcut is now justified: reverse the sign of the non-time-derivative term. The electromagnetic energy density is

$$T^{00}_{\mathrm{EM}} = \mathcal{H}_{\mathrm{EM}} = \frac{1}{2}\left(\mathbf{E}^2 + \mathbf{B}^2\right), \tag{1.38}$$

and

$$H_{\mathrm{EM}} = \int d^3x \, \frac{1}{2}\left(\mathbf{E}^2 + \mathbf{B}^2\right). \tag{1.39}$$

Unlike the Lagrangian density, the energy density is nonnegative.

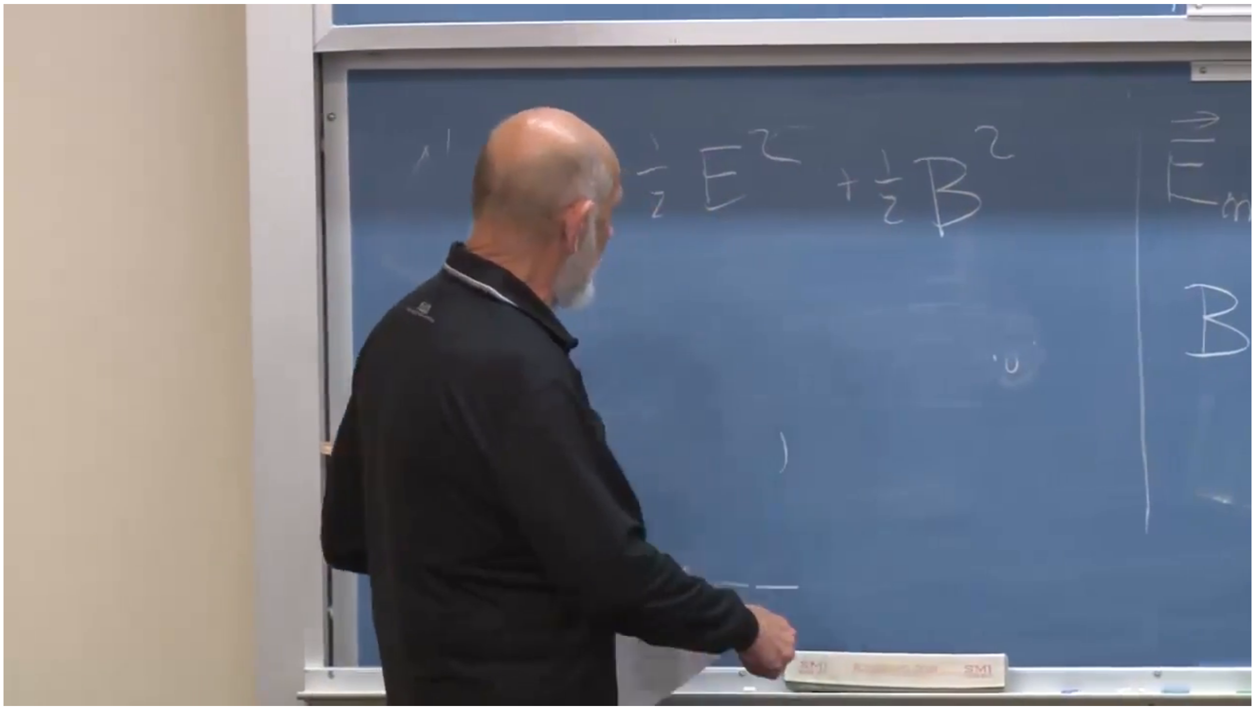

Figure 1.22: The positive electromagnetic Hamiltonian density $\frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$.
Lecture frame: 01:32:00.

For a plane wave in vacuum, $\mathbf{E} \perp \mathbf{B}$, the two fields are in phase, and in $c = 1$ units they have equal magnitude. Electric and magnetic contributions therefore divide the wave energy equally:

$$u_E = \frac{1}{2}\mathbf{E}^2, \qquad u_B = \frac{1}{2}\mathbf{B}^2, \qquad u_E = u_B. \tag{1.40}$$

1.10 Deriving Electromagnetic Momentum

Noether's translation formula must now be summed over the three fields A_m. With the translation sign chosen so that the physical momentum points along wave propagation, the n-component is

$$P_n = \int d^3x\, E_m\, \partial_n A_m, \tag{1.41}$$

where m is summed. The lecture deliberately pauses over the sign: it depends on the earlier electric-field and active/passive translation choices. The gauge-invariant physical result will fix it.

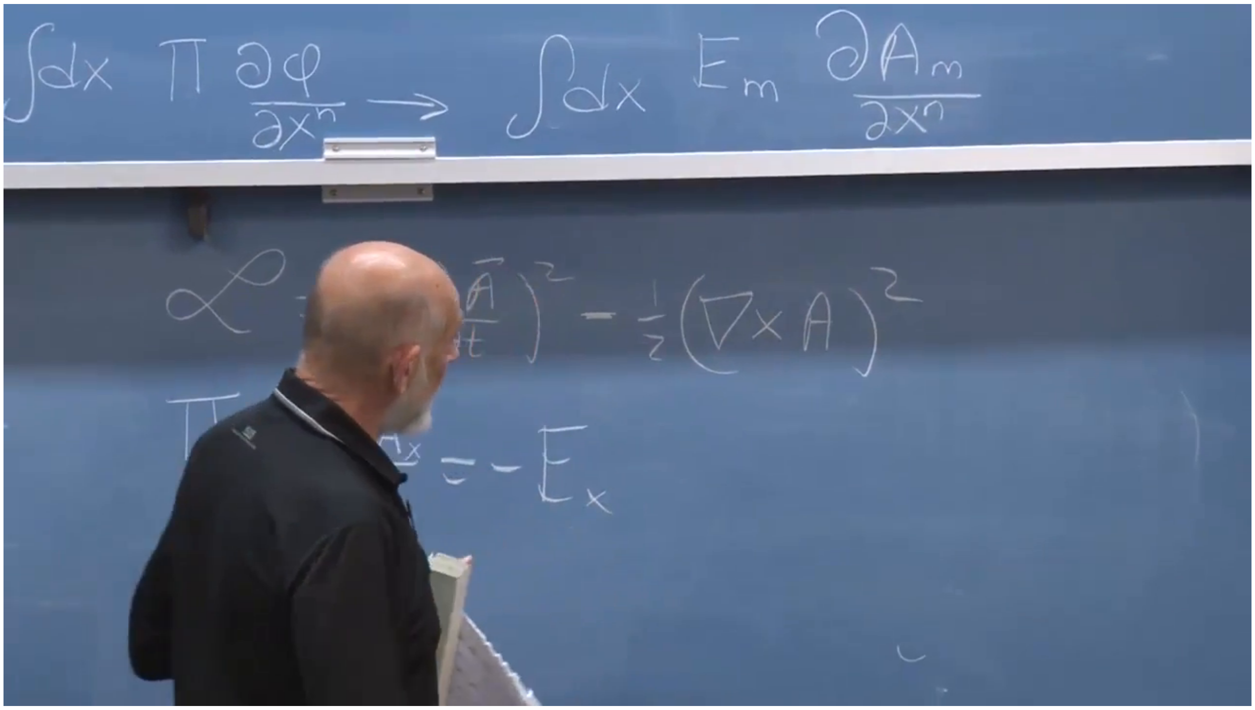

Figure 1.23: The scalar-field Noether formula promoted to the three components of the vector potential.

Lecture frame: 01:36:40.

The integrand still contains the gauge potential rather than only the physical fields. Insert and subtract $E_m \partial_m A_n$:

$$E_m \partial_n A_m = E_m \left(\partial_n A_m - \partial_m A_n\right) + E_m \partial_m A_n. \tag{1.42}$$

The first term is a cross product,

$$E_m \left(\partial_n A_m - \partial_m A_n\right) = (\mathbf{E} \times \mathbf{B})_n. \tag{1.43}$$

The second term integrates by parts:

$$\int d^3x\, E_m \partial_m A_n = \int_{\partial V} d\Sigma_m\, E_m A_n - \int d^3x\, A_n \partial_m E_m. \tag{1.44}$$

If the fields fall off so that the boundary term vanishes, and if the region is source free so that $\boldsymbol{\nabla} \cdot \mathbf{E} = 0$, the second term contributes nothing.

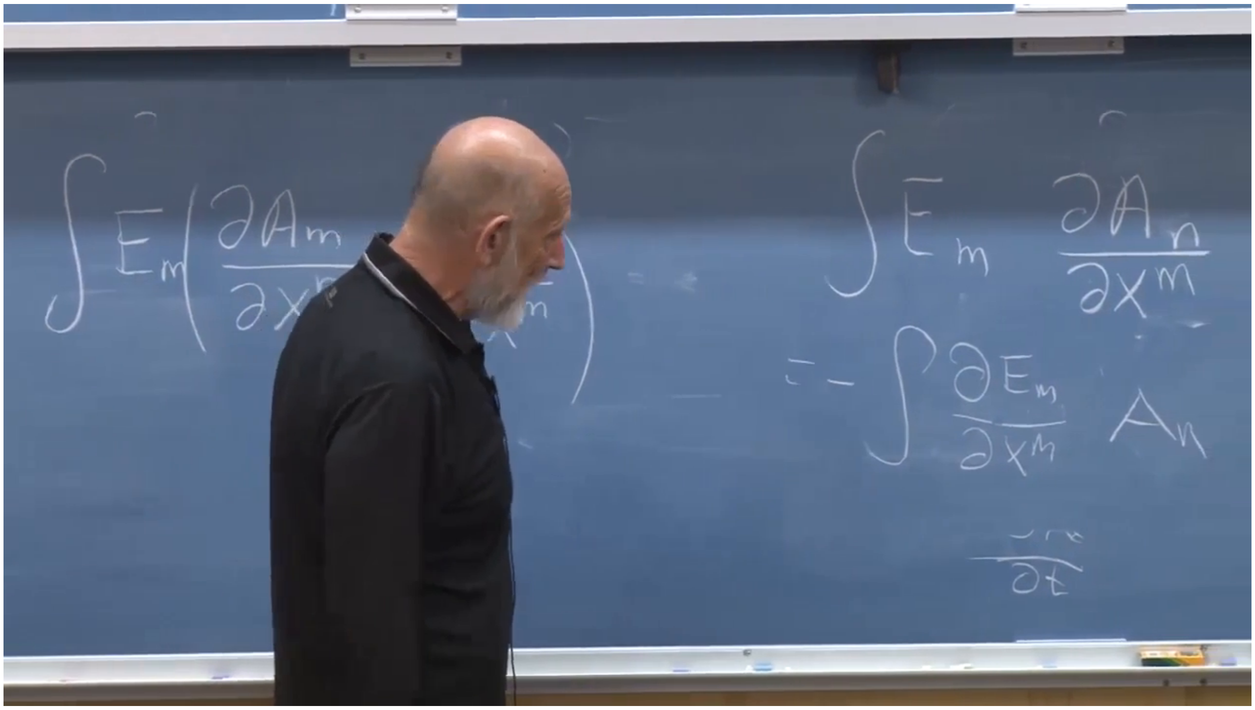

Figure 1.24: The added term transferred by integration by parts to $\nabla \cdot \mathbf{E}$.
Lecture frame: 01:40:00.

The total field momentum is therefore

$$\mathbf{P}_{\mathrm{EM}} = \int d^3x\, \mathbf{E} \times \mathbf{B}, \qquad \mathbf{g}_{\mathrm{EM}} = \mathbf{E} \times \mathbf{B}, \tag{1.45}$$

in the units of the lecture. The vector $\mathbf{g}_{\mathrm{EM}}$ is the electromagnetic momentum density.

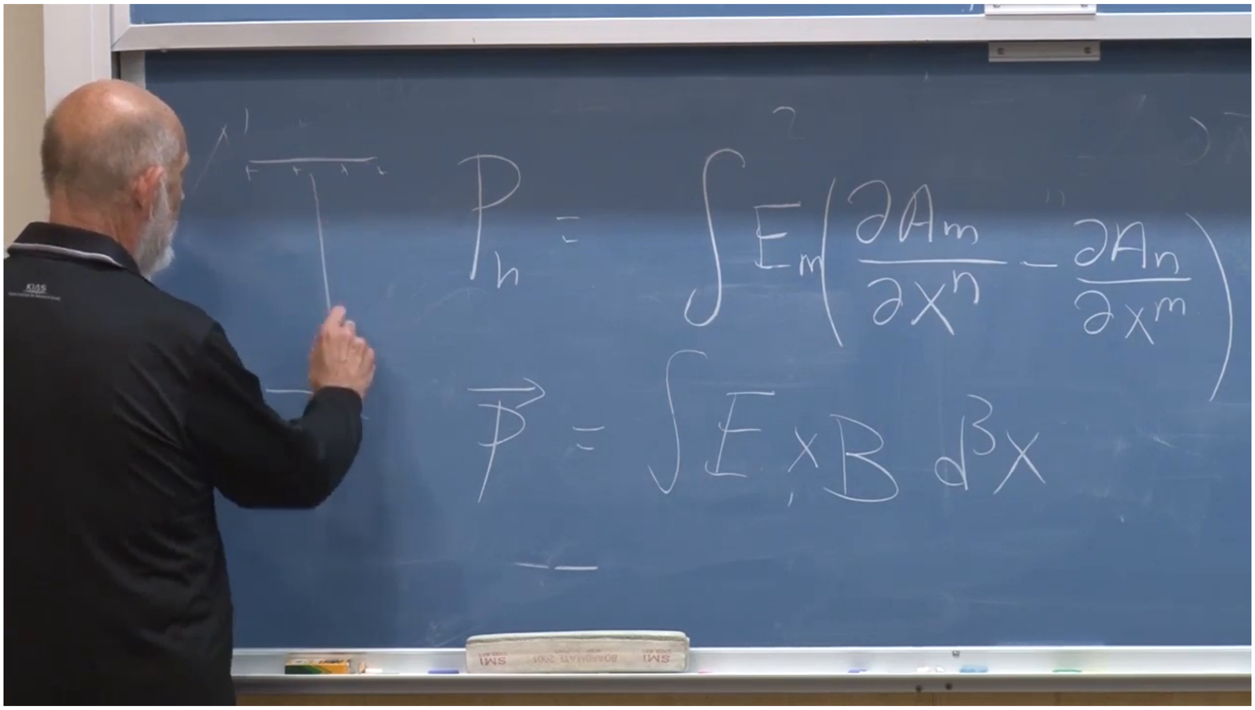

Figure 1.25: The gauge-potential expression reduced to the gauge-invariant momentum $\int d^3x \, \mathbf{E} \times \mathbf{B}$. *Lecture frame: 01:41:10.*

Question from the audience. Where was $\nabla \cdot \mathbf{E} = 0$ used, and what if charges are present?

Response. It removes the volume term after integration by parts, so the displayed field-only derivation applies to a free electromagnetic wave. With charge present, that term is not zero and the momentum of the charged matter must be included. Translation invariance and momentum conservation concern the complete matter–field system. [a]

[a] Lecture timestamp: 01:41:55.

1.10.1 Direction of wave momentum

In a plane wave, $\mathbf{E}$, $\mathbf{B}$, and the propagation direction form a right-handed triad. Half a wavelength later both $\mathbf{E}$ and $\mathbf{B}$ reverse:

$$(-\mathbf{E}) \times (-\mathbf{B}) = \mathbf{E} \times \mathbf{B}. \tag{1.46}$$

The momentum density therefore points in one fixed direction throughout the oscillation, the direction in which the wave travels.

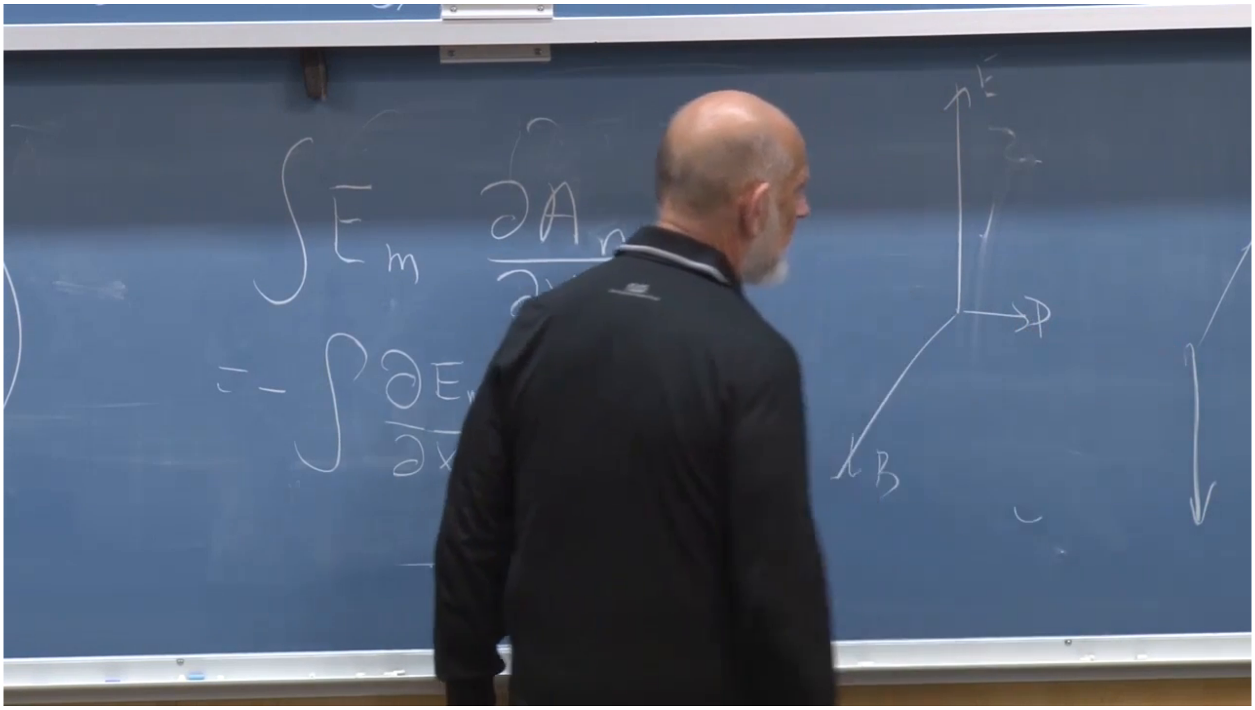

Figure 1.26: The plane-wave geometry: electric and magnetic fields reverse together, while their cross product continues along the propagation axis.

Lecture frame: 01:43:10.

This is why light pressure is real. Absorption transfers the field momentum to matter; reflection reverses the light momentum and can transfer twice the normal component to the reflector.

1.11 Where the Potential Meets the Observable Fields

At the close, the class asks for the points of contact between A_μ and the observable fields. They are precisely

$$\begin{aligned} E_n &= -\partial_t A_n + \partial_n A_0, \\ B_n &= \epsilon_{nmk} \partial_m A_k. \end{aligned} \tag{1.47}$$

Temporal gauge removes the A_0 term but does not change the physical fields.

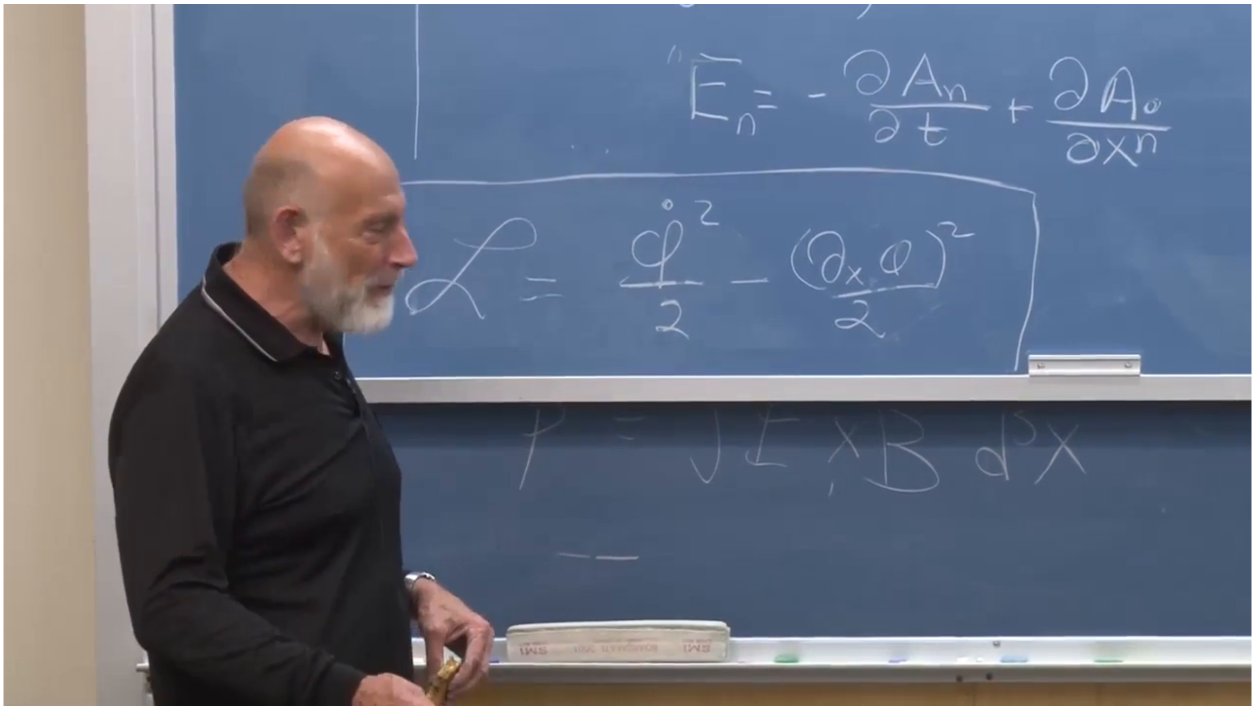

Figure 1.27: The general electric-field relation restored after the temporal-gauge calculation, together with the covariant scalar Lagrangian recalled below it.

Lecture frame: 01:45:40.

Question from the audience. Exactly where do the vector potential and the electric and magnetic fields connect?

Response. The electric field combines the time derivative of the spatial potential with the spatial derivative of A_0; the magnetic field is the curl of the spatial potential. In temporal gauge this becomes simply $\mathbf{E} = -\dot{\mathbf{A}}$ and $\mathbf{B} = \nabla \times \mathbf{A}$. [a]

[a]Lecture timestamp: 01:44:15.

Question from the audience. Is the potential more fundamental than **E** and **B**?

Response. The action and gauge principle are most naturally formulated with A_μ, which makes the potential structurally fundamental. Yet A_μ is not unique: gauge-related potentials represent the same physics, and the local fields are gauge invariant. The two statements are compatible. Fundamental variables need not themselves be directly observable. [a]

[a]Lecture timestamp: 01:46:36.

Question from the audience. Can the gauge potential itself exert a force?

Response. A gauge-dependent representative cannot produce a gauge-dependent measurable force. Charged matter is coupled through the potential in the action, but the classical Lorentz force can be written entirely in terms of the gauge-invariant electric and magnetic fields. [a]

[a]Lecture timestamp: 01:47:49.

Question from the audience. If crossed electric and magnetic fields are static, must their momentum be zero?

Response. Not necessarily for the field by itself: $\mathbf{E} \times \mathbf{B}$ can be nonzero even when both fields are time independent. In a closed static apparatus the complete momentum budget also includes the charges, currents, supports, and any mechanical or hidden momentum. The brief classroom answer that the total must balance is the correct conservation statement; setting the field contribution alone to zero would be too strong. [a]

[a] Lecture timestamp: 01:48:01.

1.12 One Tensor, Two Meanings of the Poynting Vector

The energy–momentum tensor organizes all the densities and fluxes:

$$\begin{aligned} &T^{00} = \text{energy density}, && T^{0i} = i\text{-momentum density}, \\ &T^{i0} = \text{energy flux in the } i\text{-direction}, && T^{ij} = j\text{-momentum flux in the } i\text{-direction}. \end{aligned} \tag{1.48}$$

For example, T^{31} is the amount of x-momentum crossing a unit area whose normal points in the z-direction per unit time. The spatial block T^{ij} is the stress tensor.

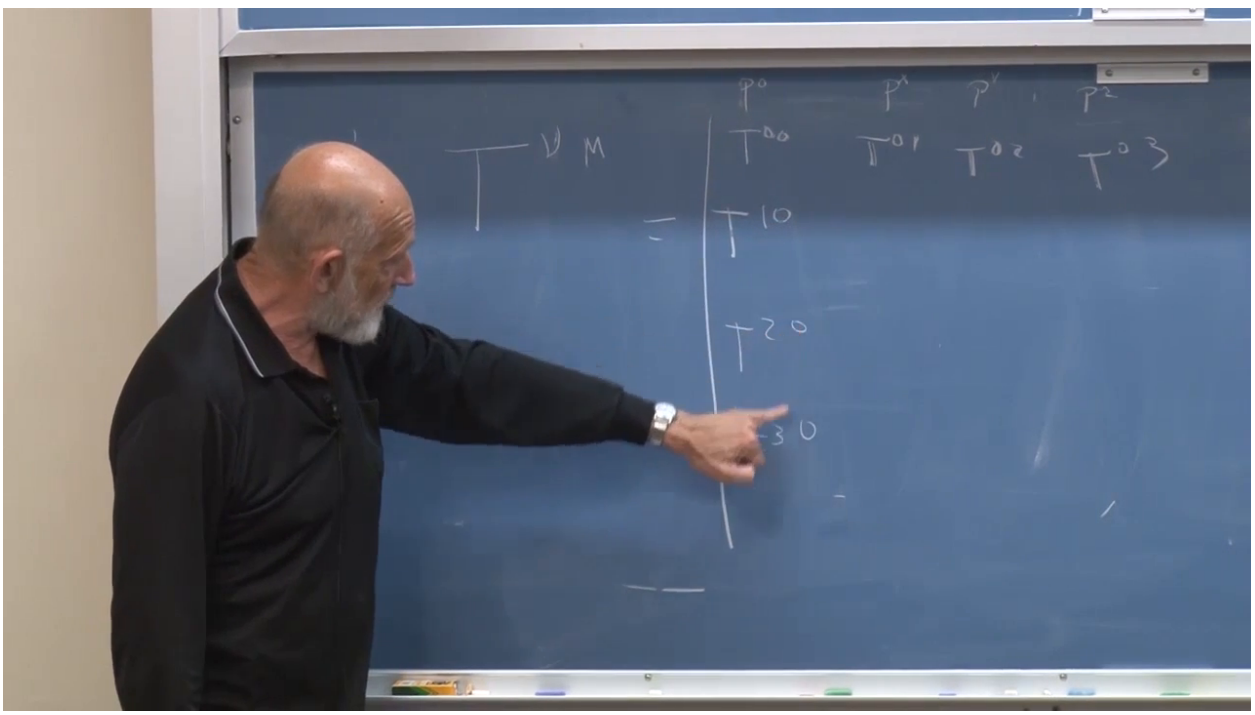

Figure 1.28: The board begins arranging density and flux components into the energy–momentum tensor.

Lecture frame: 01:51:10.

For the relativistic systems under discussion, the physical energy–momentum tensor can be chosen symmetric:

$$T^{\mu\nu} = T^{\nu\mu}. \tag{1.49}$$

Consequently,

$$T^{0i} = T^{i0}. \tag{1.50}$$

The density of i-momentum equals the flux of energy in the i-direction when $c = 1$. This is the answer to a final audience memory: the Poynting vector is indeed a power density.

Question from the audience. Is $\mathbf{E}\times\mathbf{B}$ the power density rather than the momentum density?

Response. It is both. Symmetry of $T^{\mu\nu}$ identifies the momentum density T^{0i} with the energy flux T^{i0}. Thus $\mathbf{S} = \mathbf{E}\times\mathbf{B}$ is energy per unit time per unit area and, in $c = 1$ units, momentum per unit volume. With ordinary units restored, $\mathbf{g} = \mathbf{S}/c^2$. [a]

[a] Lecture timestamp: 01:48:20.

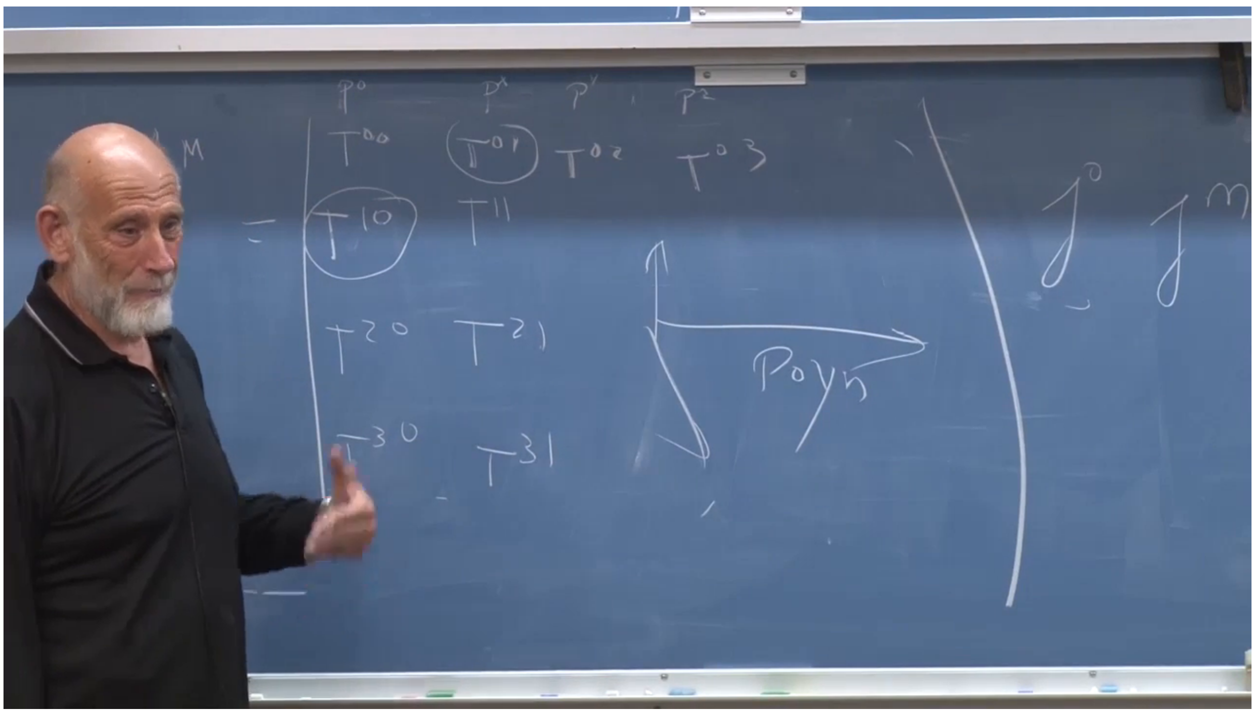

Figure 1.29: The symmetric tensor relation equating momentum density with energy flux, labeled on the board by the Poynting vector.

Lecture frame: 01:53:50.

1.13 The Bridge to General Relativity

The course closes by bringing classical mechanics, field theory, and electromagnetism into one structure:

1. Time-translation invariance gives the Hamiltonian and energy.
2. Spatial-translation invariance gives Noether momentum.
3. Replacing discrete coordinates by spatially labeled fields turns sums into integrals and conserved quantities into local densities.

4. Gauge fixing exposes the mechanical coordinates of the Maxwell field without changing its observables.
5. The Maxwell Hamiltonian gives $\frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$, while spatial translations give $\mathbf{E} \times \mathbf{B}$.
6. The symmetric energy–momentum tensor unifies energy density, momentum density, energy flux, and stress.

This is the quantity needed at the next stage. In general relativity, the energy–momentum tensor appears as the source in Einstein's field equation. The final lecture has therefore not merely appended an electromagnetic calculation; it has prepared the language in which matter and fields tell spacetime how to curve.